

 rcfdtools

 General update.

b7ee94f · 2 months ago



1304 lines (878 loc) · 776 KB

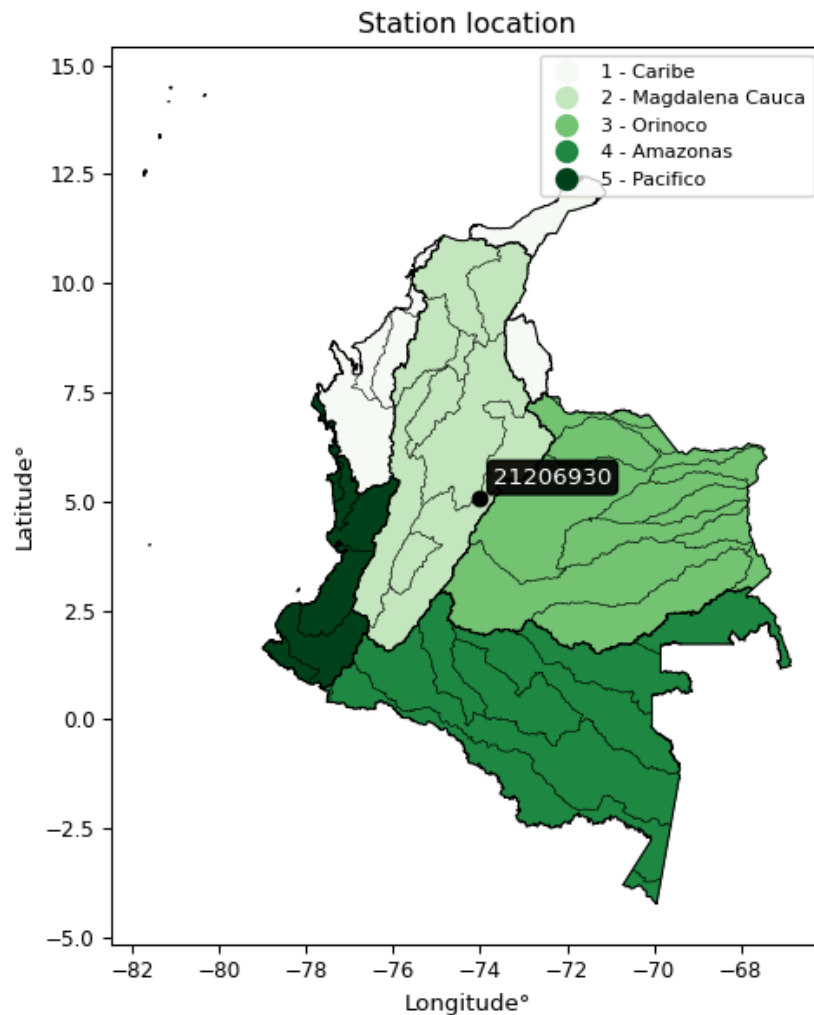
R.HydroTools

PMax24h Station (Rain): 21206930

Maximum Precipitation in 24 hours (PMax24h) is the greatest amount of rainfall for a specific duration that is meteorologically possible for a given location, acting as a "worst-case" scenario for extreme storms, crucial for designing safety-critical infrastructure like bridges, river deviations, dams, spillways, and nuclear plants to prevent catastrophic failure. The PMax24h is related with the Probable Maximum Precipitation (PMP) and it is calculated by hydrologists using meteorological data to determine the upper limit of extreme rainfall, often leading to the Probable Maximum Flood (PMF) for flood control design, and is increasingly being studied for climate change impacts. Probable Maximum Precipitation (PMP) is the theoretical upper limit of rainfall, a deterministic estimate for extreme events, while probability distributions (like GEV, Gumbel) describe the likelihood and frequency of various precipitation amounts, including rare ones, showing how often events occur, with PMP representing the extreme end of these distributions, used for critical infrastructure design to ensure safety against the worst conceivable weather, unlike standard statistical forecasts which cover typical probabilities.

The most common probability distributions in hydrology, used for analyzing floods, rainfall, and streamflow, include the Normal, Log-Normal, Gumbel, Gamma (including Log-Pearson Type III), and Generalized Extreme Value (GEV) distributions, often chosen based on the data's skewness and whether modeling extremes or general conditions, with Gumbel and GEV popular for extreme events like floods, while the Normal distribution serves as a baseline, though often requiring transformations for skewed hydrological data. These distributions help in designing water infrastructure, managing water resources, and forecasting hydrologic events.

Hydrological data often isn't perfectly normal (it's skewed), so different distributions are needed for different applications, such as: **Flood Frequency Analysis**: Using Gumbel, GEV, or Log-Pearson Type III for predicting extreme flood magnitudes and their return periods, **Rainfall Analysis**: Normal for annual totals, but Log-Normal, Weibull, or GEV for intensity or extreme daily rainfall, **Streamflow Modeling**: Gamma and Log-Normal for maximum flows, GEV for minimum flows, and Kappa for daily flows.



A. General information

1. General running parameters

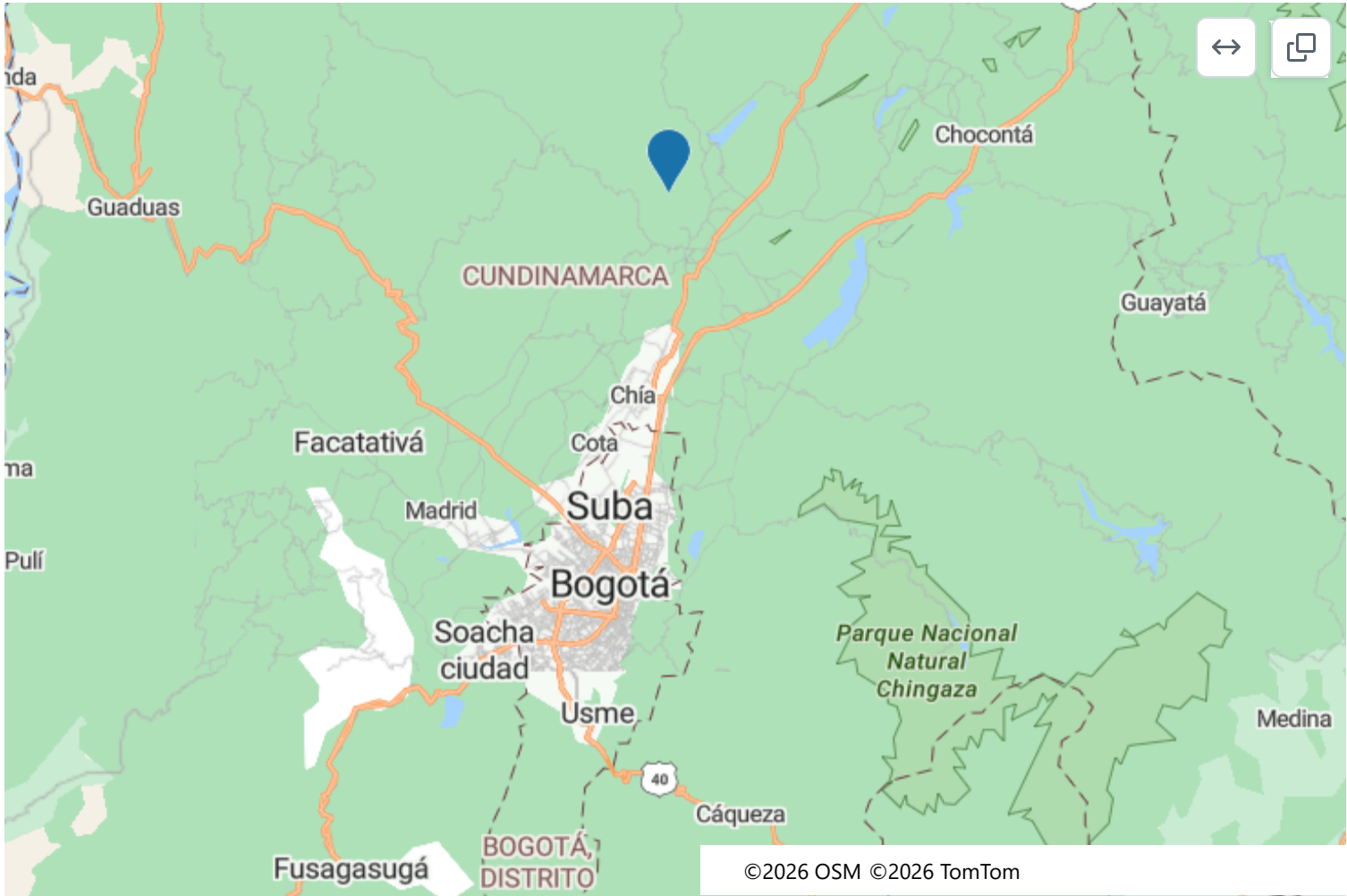
• app_version: v20260312. • runtime: 2026-03-12 07:15:23.203550. • Python version: 3.14.0. • SciPy version: 1.16.3. • Pandas version: 2.3.3. • NumPy version: 2.3.5. • Stations dataset (station_dataset_file): dataset/pmax24h_in/automatic_colombia_2003_2025.csv. • Stations catalog (station_catalog_file): dataset/CNE.xls. • Minimum year to eval til year_max (date_min): 1900. • Maximum year to eval since year_min (date_max): 2025. • Creates, save and include plots into reports (create_plot): True. • Plot only fit distributions with $\Delta o > \Delta$ (plot_only_fit): True. • Plot only simple graphs avoiding multiple CDFs and multiple Extreme values plots (plot_only_simple): True. • Eval low extreme values, if False, evaluates high extreme values (low_extreme): False. • Eval every SciPy distribution as logarithmic (pdist_logarithmic_on): True. • Standard deviation normalized (ddof): 1.0. • Return periods to eval in years (Tr): [2, 2.33, 5, 10, 15, 20, 25, 50, 75, 100, 200, 250, 500, 750, 1000]. • Minimum data sample per station, 0 means any (minimum_sample): 5. • Z-Score maximum threshold to adjust a value, 0 means disable (zscore_max): 3.5. • Z-Score minimum threshold to adjust a value, 0 means disable (zscore_min): -2.5. • Avoid zeros, e.g. rain = 0 (avoid_zeros): True. • Avoid null values (avoid_nans): True.

Table: dataset.csv

Delta Degrees of Freedom (DDOF): is a parameter used in the formulas for calculating variance and standard deviation. When ddof=0, the divisor is N (the total number of observations) and this is used when your data set is the entire population. When ddof=1, the divisor is $N - 1$ and this is used when your data set is a sample drawn from a larger population. Using $N - 1$ provides an unbiased estimate of the population variance (known as Bessel's correction).

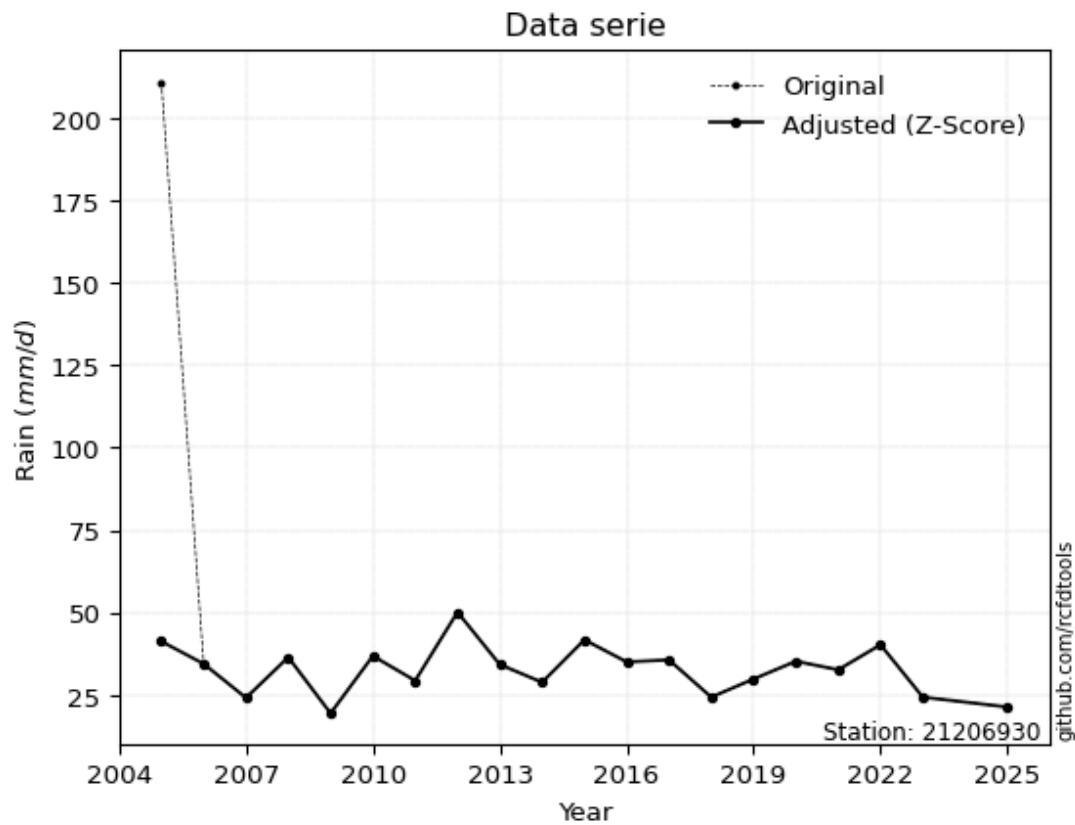
2. Station info and location

A	ENTIDAD	AREA_HIDROGRAFICA	ZONA_HIDROGRAFICA	SUBZONA_HIDROGRAFICA
I	INSTITUTO DE HIDROLOGIA METEOROLOGIA Y ESTUDIOS AMBIENTALES	Magdalena Cauca	Alto Magdalena	Río Bogotá



3. Discrete values table and plot

	0	1	2	3	4	5	
Date	2005	2006	2007	2008	2009	2010	2011
Value	41.225	34.4	24.1	36.4	19.4	36.8	29.1
Value_initial	211	34.4	24.1	36.4	19.4	36.8	29.1
count	20	20	20	20	20	20	20
mean	41.225	41.225	41.225	41.225	41.225	41.225	41.225
std	40.6462	40.6462	40.6462	40.6462	40.6462	40.6462	40.6462
zscore	4.1769	-0.167913	-0.421319	-0.118707	-0.536951	-0.108866	-0.29338



The initial value (value_initial) correspond to the initial obtained or null completed value, and value (value) is adjusted only when the initial value is outside the valid range (outlier) definite through the Z-Score value. If Z-Score is active, values out of range or outliers are replaced with the station mean value. A Z-Score (or standard score) measures how many standard deviations a data point is from the mean of a dataset, indicating its relative position; positive scores mean above the mean, negative mean below, and a score of zero is exactly at the mean, allowing for comparison of values from different distributions. It's calculated using the formula $z = (x - \mu) / \sigma$, where x is the data point, μ is the mean, and σ is the standard deviation, helping identify outliers and understand probability.

Note: In this study, "completing" (filling gaps in) and "extending" (forecasting or lengthening the historical record) rainfall data series are avoided for certain critical analyses because they introduce artificial bias and uncertainty, which can distort the statistics of rare events (extremes), alter day-to-day variability, and compromise the integrity and homogeneity of the original data. Some reasons against Completing/Extending rainfall data are: **Distortion of Extremes and Variability:** The primary issue is that most gap-filling or extension methods rely on averaging or regression with nearby stations, which inherently smooths the data. Rainfall is highly variable in space and time, with a large percentage of zero values and occasional large storms. Infilling methods often underestimate the highest values and overestimate the lowest (zero) values, which severely distorts analyses focused on flood frequency, drought severity, and intensity-duration-frequency (IDF) curves, **Loss of Data Homogeneity:** Infilled or extended data are synthetic estimates, not actual measurements. Combining these estimates with observed data can violate the assumption of data homogeneity, which requires that the data properties do not change over time. Inhomogeneity can lead to incorrect conclusions about long-term trends or changes in climate patterns, **Propagation of Errors:** Errors and biases from the original data (e.g., wind-induced undercatch, instrument errors, site changes) or the data used for imputation can propagate through the analysis, leading to unreliable hydrological models and inaccurate predictions, **Compromised Statistical Integrity:** For statistical analyses, especially those involving the frequency of events, using artificial data can introduce time bias or autocorrelation that doesn't exist in reality, making it difficult to assess the true probability of events, **Difficulty in Validation:** It is difficult to accurately validate the quality of filled or extended data, especially for past periods where ground-truth measurements are unavailable. The reliability of imputation techniques heavily depends on the density of the surrounding station network and the complexity of the local topography.

4. Active continuous probability distributions from SciPy (80 of 104 available)

A Probability Density Function (PDF) describes the relative likelihood of a continuous random variable falling within a specific range, where the total area under its curve equals 1, and the area over any interval gives the actual probability for that range. Unlike discrete probabilities (like rolling a die), a PDF shows density, not direct probability for a single point (which is zero), with higher points indicating higher likelihood, often visualized as a bell curve for normal distributions.

A log of a probability density function (log-PDF) is simply the logarithm of the PDF's value, denoted as $\log(f(x))$, useful for converting multiplication of probabilities into addition, improving numerical stability with tiny numbers, and connecting to information theory concepts like entropy. Instead of dealing with very small probabilities (e.g., 0.000001), log-PDFs use negative numbers (e.g., $\log(0.000001) \approx -11.51$) that are easier for computers to handle, preventing underflow errors and simplifying complex calculations. In this study, each active probability distribution from SciPy is also evaluated in the *log* form.

[scipy.stats](#) is a Python's powerful submodule within the SciPy library for comprehensive statistical analysis, offering over 130 probability distributions (like Normal, Poisson, etc.), functions for descriptive stats (mean, variance), hypothesis testing (t-tests, chi-square), random variable generation, and statistical tests, making it essential for data science, modeling, and research. It provides tools to explore, model, and draw conclusions from data efficiently, working seamlessly with [NumPy](#).

id	p_dist	n_parameter	fit_method	label	active	ref
0	alpha	3	MLE	Alpha	True	
1	anglit	2	MM	Anglit	True	

id	p_dist	n_parameter	fit_method	label	active	ref
2	arcsine	2	MM	Arcsine	True	
3	argus	3	MLE	Argus	True	
4	beta	4	MLE	Beta	True	
5	betaprime	4	MLE	Beta prime	True	
6	bradford	3	MLE	Bradford	True	
7	burr	4	MLE	Burr (Type III)	True	
8	burr12	4	MLE	Burr (Type III) 12	True	
9	cosine	2	MLE	Cosine	True	
10	crystalball	4	MLE	Crystalball	True	
11	dgamma	3	MLE	Double gamma	True	
12	dweibull	3	MLE	Double Weibull	True	
13	erlang	3	MLE	Erlang	True	
14	exponnorm	3	MLE	Exponentially modified Normal	True	
15	exponpow	3	MLE	Exponential power	True	
16	exponweib	4	MLE	Exponentiated Weibull	True	
17	f	4	MLE	F	True	
18	fatiguelife	3	MLE	Fatigue-life (Birnbaum-Saunders)	True	
19	foldnorm	3	MM	Fold Normal	True	
20	gamma	3	MLE	Gamma	True	
21	gausshyper	6	MLE	Gauss hypergeometric	True	
22	genexpon	5	MLE	Generalized exponential	True	
23	genextreme	3	MLE	Generalized exponential	True	
24	gengamma	4	MLE	Generalized gamma	True	
25	genhalflogistic	3	MLE	Generalized half-logistic	True	
26	genlogistic	3	MLE	Generalized logistic	True	

id	p_dist	n_parameter	fit_method	label	active	ref
27	gennorm	3	MLE	Generalized Normal	True	🎓
28	genpareto	3	MLE	Generalized Pareto	True	🎓
29	gompertz	3	MLE	Gompertz (or truncated Gumbel)	True	🎓
30	gumbel_l	2	MM	Gumbel Left Skew	True	🎓
31	gumbel_r	2	MM	Gumbel Right Skew	True	🎓
32	halfgennorm	3	MLE	Upper half of a generalized normal	True	🎓
33	halflogistic	2	MM	Half-logistic	True	🎓
34	halfnorm	2	MM	Half Normal	True	🎓
35	hypsecant	2	MM	hyperbolic secant	True	🎓
36	invgamma	3	MLE	Inverted gamma	True	🎓
37	invgauss	3	MLE	Inverse Gaussian	True	🎓
38	invweibull	3	MLE	Inverted Weibull	True	🎓
39	johnsonsb	4	MLE	Johnson SB	True	🎓
40	johnsonsu	4	MLE	Johnson Su	True	🎓
41	kappa3	3	MLE	Kappa 3	True	🎓
42	kappa4	4	MLE	Kappa 4	True	🎓
43	ksone	3	MLE	Kolmogorov-Smirnov one-sided test statistic distribution	True	🎓
44	kstwobign	2	MLE	Limiting distribution of scaled Kolmogorov-Smirnov two-sided test statistic	True	🎓
45	laplace	2	MM	Laplace	True	🎓
46	levy_l	2	MLE	Left-skewed Levy	True	🎓
47	loggamma	3	MLE	Log gamma	True	🎓

id	p_dist	n_parameter	fit_method	label	active	ref
48	logistic	2	MM	Logistic (or Sech-squared)	True	🎓
49	loglaplace	3	MLE	Log-Laplace	True	🎓
50	lomax	3	MLE	Lomax (Pareto of the second kind)	True	🎓
51	maxwell	2	MM	Maxwell	True	🎓
52	mielke	4	MLE	Mielke Beta-Kappa / Dagum	True	🎓
53	moyal	2	MM	Moyal	True	🎓
54	nakagami	3	MLE	Nakagami	True	🎓
55	ncf	5	MLE	Non-central F distribution	True	🎓
56	nct	4	MLE	Non-central Student's t	True	🎓
57	norm	2	MM	Normal	True	🎓
58	pearson3	3	MM	Pearson type III	True	🎓
59	powerlaw	3	MLE	Power-function	True	🎓
60	rayleigh	2	MM	Rayleigh	True	🎓
61	rdist	3	MLE	R-distributed (symmetric beta)	True	🎓
62	rice	3	MLE	Rice	True	🎓
63	semicircular	2	MM	Semicircular	True	🎓
64	skewnorm	3	MLE	Skew normal	True	🎓
65	t	3	MLE	Student's t	True	🎓
66	trapezoid	4	MLE	Trapezoid	True	🎓
67	triang	3	MLE	Triangular	True	🎓
68	truncexpon	3	MLE	Truncated exponential	True	🎓
69	truncnorm	4	MLE	Truncated normal	True	🎓
70	truncpareto	4	MLE	Upper truncated Pareto	True	🎓

id	p_dist	n_parameter	fit_method	label	active	ref
71	truncweibull_min	5	MLE	Doubly truncated Weibull minimum	True	🎓
72	tukeylambda	3	MLE	Tukey-Lambda	True	🎓
73	uniform	2	MLE	Uniform	True	🎓
74	vonmises	3	MLE	Von Mises	True	🎓
75	vonmises_line	3	MLE	Von Mises line	True	🎓
76	wald	2	MM	Wald	True	🎓
77	weibull_max	3	MLE	Weibull maximum	True	🎓
78	weibull_min	3	MLE	Weibull minimum	True	🎓
79	wrapcauchy	3	MLE	Wrapped Cauchy	True	🎓

n_parameter: # arguments & localization & scale.

Fit methods: (MLE) maximum likelihood, (MM) L-moments.

Inactive (With the only purpose of prevent loops, zero divisions, infinite values, over high estimated values or horizontal trending for recurrence intervals estimations, the follow distributions were disabled.): lognorm, norminvgauss, powernorm, powerlognorm, cauchy, halfcauchy, foldcauchy, skewcauchy, chi2, expon, fisk, genhyperbolic, geninvgauss, gibrat, kstwo, laplace_asymmetric, levy, levy_stable, ncx2, pareto, rel_breitwigner, recipinvgauss, studentized_range, loguniform,

B. Probability (PDF) vs. Empirical (EDF) distributions functions

A continuous probability distribution (CPD) describes probabilities for variables that can take any value within a range (like rain any time), unlike discrete variables with specific outcomes (like temperature). It uses a Probability Density Function (PDF), a curve where the total area under it equals 1, and the probability of the variable falling within an interval (a to b) is found by calculating the area under the curve between those points. A key feature is that the probability of hitting any single exact value is zero, so probabilities are always expressed for ranges, e.g., $P(a \leq X \leq b)$.

Distributions parameters from station values

	station	p_dist	n	loc	scale	shape
0	21206930	alpha	20	-117.066	2983.2	19.9701948431228
1	21206930	anglit	20	32.7362	21.9441	

	station	p_dist	n	loc	scale	shape
2	21206930	arcsine	20	22.1279	21.2167	
3	21206930	argus	20	13.1667	37.523	2.42515789492900
4	21206930	beta	20	19.4	30.7337	0.76886863659282
5	21206930	betaprime	20	-0.568686	37.6236	35.8564699253168
6	21206930	bradford	20	19.4	30.7001	1.07747420016540
7	21206930	burr	20	-0.164049	37.87	11.6891055878164
8	21206930	burr12	20	0.782645	42.7449	5.48799801296916
9	21206930	cosine	20	33.2215	6.6978	
10	21206930	crystalball	20	32.7363	7.50123	1891.29952384011
11	21206930	dgamma	20	33.1118	3.74959	1.60952625230881
12	21206930	dweibull	20	34.4	5.85833	0.98047443005868
13	21206930	erlang	20	-31.5738	0.877867	73.2572316223049
14	21206930	exponnorm	20	29.883	6.94218	0.41101000608495
15	21206930	exponpow	20	19.4	1.53904	0.14571132661885
16	21206930	exponweib	20	-13.0128	1.94541	5211.99282283881
17	21206930	f	20	-33.4279	66.146	158.519057334316
18	21206930	fatiguelife	20	-70.7474	103.212	0.07252609619918
19	21206930	foldnorm	20	32.5058	1.85599	4.44635842240939
20	21206930	gamma	20	-32.8727	0.860267	76.2619363411872
21	21206930	gausshyper	20	19.4	30.9704	0.97317405930713
22	21206930	genexpon	20	19.3989	5.39003	0.11523883134309
23	21206930	genextreme	20	29.9	7.25332	0.22777440195297
24	21206930	gengamma	20	19.4	28.1498	0.16455310373363
25	21206930	genhalflogistic	20	19.4	14.8502	0.46297197882462
26	21206930	genlogistic	20	31.9195	4.51984	1.13155396180165
27	21206930	gennorm	20	32.6635	11.2749	2.28310512512923

	station	p_dist	n	loc	scale	shape
28	21206930	genpareto	20	-0.154961	64.8845	-1.2911064371520
29	21206930	gompertz	20	24.1	2.23129	0.00014702675843
30	21206930	gumbel_l	20	36.1122	5.84869	
31	21206930	gumbel_r	20	29.3603	5.84869	
32	21206930	halfgennorm	20	19.4	1.30358	0.42294119963162
33	21206930	halflogistic	20	23.8456	6.41327	
34	21206930	halfnorm	20	22.8075	12.4438	
35	21206930	hypsecant	20	32.7363	4.77543	
36	21206930	invgamma	20	-50.3965	10115.9	122.640754912837
37	21206930	invgauss	20	-9.52492	1266.82	0.03337478220684
38	21206930	invweibull	20	-3.52516e+08	3.52516e+08	52085167.8590360
39	21206930	johnsonsb	20	15.4517	41.282	0.43916317239446
40	21206930	johnsonsu	20	2.01496e+07	1.80017e+08	2696620.23097464
41	21206930	kappa3	20	19.4	18.6051	8.89039809200263
42	21206930	kappa4	20	15.8353	35.0705	1.11329399705417
43	21206930	ksone	20	16.33	36.84	1.0
44	21206930	kstwobign	20	4.63307	32.1788	
45	21206930	laplace	20	32.7363	5.30417	
46	21206930	levy_l	20	51.8142	12.1396	
47	21206930	loggamma	20	-2072.32	289.017	1456.56436166764
48	21206930	logistic	20	32.7363	4.13565	
49	21206930	loglaplace	20	-0.0948054	34.4948	5.35255351934014
50	21206930	lomax	20	19.4	250120	18721.9588658293
51	21206930	maxwell	20	14.9615	11.1387	
52	21206930	mielke	20	-0.193856	37.8957	4.71648987137723
53	21206930	moyal	20	28.4466	3.37674	

	station	p_dist	n	loc	scale	shape
54	21206930	nakagami	20	16.7824	17.6293	1.13080079539429
55	21206930	ncf	20	-0.747898	26.472	37.5883245512565
56	21206930	nct	20	-43.2665	6.14829	161.183619178038
57	21206930	norm	20	32.7363	7.50123	
58	21206930	pearson3	20	32.7363	7.50122	0.15608363308185
59	21206930	powerlaw	20	19.4	30.7	0.36428776034876
60	21206930	rayleigh	20	18.386	11.4499	
61	21206930	rdist	20	28.1478	8.74782	1.34725978536810
62	21206930	rice	20	16.7616	8.72182	1.44717994402638
63	21206930	semicircular	20	32.7363	15.0025	
64	21206930	skewnorm	20	27.5979	9.09237	1.00254849046592
65	21206930	t	20	32.7364	7.50239	509851.607726604
66	21206930	trapezoid	20	16.33	36.84	1.0
67	21206930	triang	20	15.2383	36.9821	0.51813399966795
68	21206930	truncexpon	20	19.4	36.288	0.84601124853639
69	21206930	truncnorm	20	32.7362	7.50123	-227.13844360732
70	21206930	truncpareto	20	19.4	2.42711e-08	0.05263154295381
71	21206930	truncweibull_min	20	19.3815	38.253	0.96502121641953
72	21206930	tukeylambda	20	30.55	12.6575	1.13519885607960
73	21206930	uniform	20	19.4	30.7	
74	21206930	vonmises	20	-2.09819	1	0.80347552928344
75	21206930	vonmises_line	20	34.75	4.88606	0.52597207187129
76	21206930	wald	20	25.235	7.50121	
77	21206930	weibull_max	20	50.1	1.39924	0.31222677106263
78	21206930	weibull_min	20	15.4319	19.5062	2.48404369624922
79	21206930	wrapcauchy	20	18.1371	5.08706	7.07880913380529

	station	p_dist	n	loc	scale	shape
80	21206930	logalpha	20	-4.29434	246.628	31.8376820201652
81	21206930	loganglit	20	3.46102	0.696415	
82	21206930	logarcsine	20	3.12438	0.673288	
83	21206930	logargus	20	2.84568	1.08772	0.00010351590968
84	21206930	logbeta	20	2.88299	1.07219	2.04376968173030
85	21206930	logbetaprime	20	-3.32986	7.06539	1251.17795697641
86	21206930	logbradford	20	2.96527	0.948749	0.00355853457896
87	21206930	logburr	20	0.0850112	3.56725	44.4550618488269
88	21206930	logburr12	20	2.18512	2.14794	6.50703413097552
89	21206930	logcosine	20	3.44894	0.206495	
90	21206930	logcrystalball	20	3.46102	0.238058	7.41507865073762
91	21206930	logdgamma	20	3.49884	0.126977	1.50220101455783
92	21206930	logdweibull	20	3.49607	0.206851	1.29465814151176
93	21206930	logerlang	20	-0.639272	0.0143401	285.846288566863
94	21206930	logexponnorm	20	3.46087	0.238058	0.00064511270483
95	21206930	logexponpow	20	2.84863	0.836583	2.15758076759410
96	21206930	logexponweib	20	2.91079	0.828692	0.27846921674238
97	21206930	logf	20	-219.329	222.789	4314422.39394578
98	21206930	logfatiguelife	20	-69.7288	73.1894	0.00324200299811
99	21206930	logfoldnorm	20	3.03089	0.25325	1.65848261532620
100	21206930	loggamma	20	-0.420716	0.0148806	260.823476535538
101	21206930	loggausshyper	20	2.96355	0.950472	1.15907480544291
102	21206930	loggenexpon	20	2.94697	0.00474284	0.00114103042724
103	21206930	loggenextreme	20	3.39094	0.25144	0.40490691569946
104	21206930	loggengamma	20	-3.47365e+06	3.47366e+06	5.54160347273023

	station	p_dist	n	loc	scale	shape
105	21206930	loggenhalflogistic	20	2.88571	1.08885	1.05887478909053
106	21206930	loggenlogistic	20	3.63026	0.0858145	0.40160026095050
107	21206930	loggennorm	20	3.43696	0.397624	3.21344237870522
108	21206930	loggenpareto	20	-0.000749569	7.4319	-1.8984254785857
109	21206930	loggompertz	20	2.96527	0.253967	0.10722930320432
110	21206930	loggumbel_l	20	3.56816	0.185613	
111	21206930	loggumbel_r	20	3.35388	0.185613	
112	21206930	loghalfgennorm	20	2.96527	0.910666	9.50687901926660
113	21206930	loghalflogistic	20	3.17887	0.20353	
114	21206930	loghalfnorm	20	3.14594	0.394899	
115	21206930	loghypsecant	20	3.46102	0.151553	
116	21206930	loginvgamma	20	-1.06554	1546.07	342.738032116136
117	21206930	loginvgauss	20	1.40098	139.993	0.01472603084493
118	21206930	loginvweibull	20	-3.34277e+07	3.34277e+07	138261288.068577
119	21206930	logjohnsonsb	20	2.72171	1.32524	-0.3151117918742
120	21206930	logjohnsonsu	20	4.74154	0.356719	11.0567840651263
121	21206930	logkappa3	20	2.96189	0.950788	855.984812697994
122	21206930	logkappa4	20	2.94268	1.01815	1.01182193044758
123	21206930	logksone	20	2.8704	1.1385	1.0
124	21206930	logkstwobign	20	2.47333	1.13039	
125	21206930	loglaplace	20	3.46102	0.168333	
126	21206930	loglevy_l	20	3.95605	0.295866	
127	21206930	logloggamma	20	2.9029	0.447609	3.96763241107314
128	21206930	loglogistic	20	3.46102	0.131248	
129	21206930	logloglaplace	20	-0.110746	3.64297	18.8794174113181
130	21206930	loglomax	20	2.96527	10.7405	21.8241433085917

	station	p_dist	n	loc	scale	shape
131	21206930	logmaxwell	20	2.89692	0.353496	
132	21206930	logmielke	20	-0.127602	3.77867	15.7550938520288
133	21206930	logmoyal	20	3.32489	0.107164	
134	21206930	lognakagami	20	-54.1859	57.6473	14587.8653542716
135	21206930	logncf	20	-3.46855	6.9261	474.837599527524
136	21206930	lognct	20	-6.13824	0.198586	2364.70485216514
137	21206930	lognorm	20	3.46102	0.238058	
138	21206930	logpearson3	20	3.46102	0.238058	-0.3645598419886
139	21206930	logpowerlaw	20	2.96527	0.948748	0.40822711796456
140	21206930	lograyleigh	20	3.00556	0.363403	
141	21206930	logrdist	20	3.36371	0.550307	1.15187320701140
142	21206930	logrice	20	2.85277	0.2577	2.10341803398496
143	21206930	logsemicircular	20	3.46102	0.476116	
144	21206930	logskewnorm	20	3.725	0.355469	-2.4856000543415
145	21206930	logt	20	3.46102	0.238055	30030278.7621253
146	21206930	logtrapezoid	20	2.8704	1.1385	1.0
147	21206930	logtriang	20	2.85618	1.12265	0.62786080436525
148	21206930	logtruncexpon	20	2.96527	1.10565	0.85809025281812
149	21206930	logtruncnorm	20	0.0447816	1.49194	1.95750208202352
150	21206930	logtruncpareto	20	-3.35544e+07	3.35544e+07	67684252.5477864
151	21206930	logtruncweibull_min	20	2.93221	1.16505	1.49700589419381
152	21206930	logtukeylambda	20	3.43978	0.5402	1.13845400497662
153	21206930	loguniform	20	2.96527	0.948748	
154	21206930	logvonmises	20	-2.82133	1	18.1150076964082
155	21206930	logvonmises_line	20	3.43965	0.150998	0.46287932735151

	station	p_dist	n	loc	scale	shape
156	21206930	logwald	20	3.22297	0.238052	
157	21206930	logweibull_max	20	4.01187	0.620902	2.46958196496675
158	21206930	logweibull_min	20	2.31609	1.23959	5.68110791963798
159	21206930	logwrapcauchy	20	2.96527	0.151002	1.55716281697374

loc (Location parameter): This shifts the distribution along the x-axis. For many common distributions, like the normal (Gaussian) distribution, loc represents the mean μ . For others, like the uniform distribution, it might represent the minimum value, or for the beta distribution, the left end of the support interval.

scale (Scale parameter): This determines the width or spread of the distribution. For the normal distribution, scale represents the standard deviation σ . For a uniform distribution, it defines the length of the interval (from *loc* to *loc* + *scale*).

shape (Shape parameters): Refers to parameters that define the specific form of a probability distribution, distinct from its location (loc) and scale (scale). These parameters are required arguments for most distribution functions. For example, a normal (Gaussian) distribution is fully defined by its location (mean) and scale (standard deviation), so it has no specific shape parameters beyond loc and scale. However, other distributions have intrinsic properties that need specification, as Gamma distribution that takes a shape parameter, often named *a* or *alpha*.

0. Cumulative distribution values - CDF (160 evaluated)

Cumulative Distribution Function (CDF), denoted as $F_X(x)$, is a function that gives the probability that a random variable X will take a value less than or equal to a specific value, x (i.e., $(P(X \geq x)$. It essentially "accumulates" probabilities from a given point up to the far right (positive infinity), providing a complete picture of the distribution's probabilities for both discrete (like rain) and continuous (like temperature) variables, helping to find probabilities over ranges or above certain values. Each station is evaluated separately ordering the x values ascending, calculating the CDF and PDF values for the activated distributions and their logarithmic forms.

CDF ordered by x ascending and calculating $\log(f(x))$ separately

id	Station	date	x	count	mean	std	zscore	Value_initial	
0	21206930	2009	19.4	20	41.225	40.6462	-0.536951	19.4	21
1	21206930	2025	21.3	20	41.225	40.6462	-0.490206	21.3	21

id	Station	date	x	count	mean	std	zscore	Value_initial	
2	21206930	2007	24.1	20	41.225	40.6462	-0.421319	24.1	21
3	21206930	2018	24.3	20	41.225	40.6462	-0.416398	24.3	21
4	21206930	2023	24.3	20	41.225	40.6462	-0.416398	24.3	21
5	21206930	2014	28.8	20	41.225	40.6462	-0.305687	28.8	21
6	21206930	2011	29.3	20	41.225	40.6462	-0.293386	29.3	21
7	21206930	2019	29.8	20	41.225	40.6462	-0.281084	29.8	21
8	21206930	2021	32.6	20	41.225	40.6462	-0.212197	32.6	21
9	21206930	2013	34.2	20	41.225	40.6462	-0.172833	34.2	21
10	21206930	2006	34.4	20	41.225	40.6462	-0.167913	34.4	21
11	21206930	2016	35	20	41.225	40.6462	-0.153151	35	21
12	21206930	2020	35.2	20	41.225	40.6462	-0.14823	35.2	21
13	21206930	2017	35.6	20	41.225	40.6462	-0.138389	35.6	21
14	21206930	2008	36.4	20	41.225	40.6462	-0.118707	36.4	21
15	21206930	2010	36.8	20	41.225	40.6462	-0.108866	36.8	21
16	21206930	2022	40.2	20	41.225	40.6462	-0.0252176	40.2	21
17	21206930	2005	41.225	20	41.225	40.6462	4.1769	211	21
18	21206930	2015	41.7	20	41.225	40.6462	0.0116862	41.7	21
19	21206930	2012	50.1	20	41.225	40.6462	0.218348	50.1	21

An Empirical Distribution Function (EDF) is a step-function estimate of a true cumulative distribution function (CDF) based on observed sample data, representing the proportion of data points less than or equal to a given value. It is calculated by ordering your data and jumping up by $1 / n$ (where n is sample size) at each unique data point, allowing analysis without assuming an underlying population distribution, and it gets closer to the true CDF as the sample size grows. For the empirical probability calculations, the parameter m correspond to the order number which means the position of the x values in an ascending order list.

The Kolmogorov-Smirnov (K-S) test is a non-parametric statistical test used to determine if a sample data set comes from a specific theoretical distribution (one-sample) or if two different samples come from the same underlying distribution (two-sample), by comparing their cumulative distribution functions (CDFs). It calculates the maximum vertical difference (D statistic) between these functions, with a small p-value indicating a significant difference and rejection of the null hypothesis (that the data/samples match).

1. EDF California (1923)

California’s estimates the true probability distribution of water-related data (like rainfall, streamflow) using observed samples, crucial for risk assessment.

$P = m / n$

1.1. Empirical values

	0	1	2	3
date	2009	2025	2007	2018
x	19.4	21.3	24.1	24.3
m	1	2	3	4
empirical_dist	edf_california	edf_california	edf_california	edf_californ
empirical	0.05	0.1	0.15	0.2
empirical_tr	1.0526315789473684	1.1111111111111112	1.1764705882352942	1.25



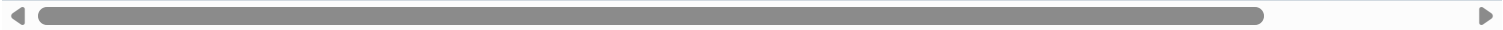
1.2. Kolmogorov-Smirnov fit test (sorted by Δ)

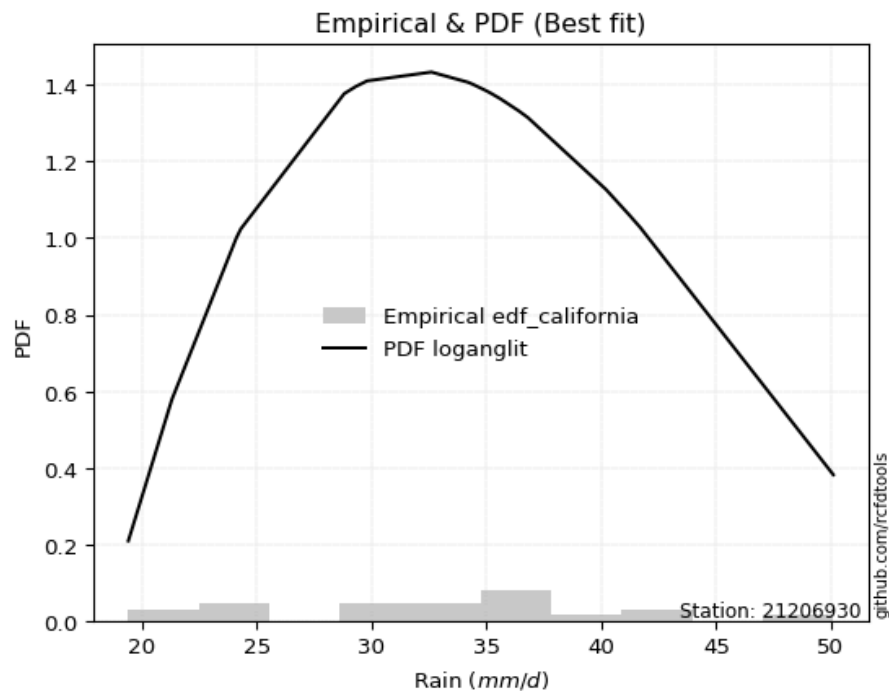
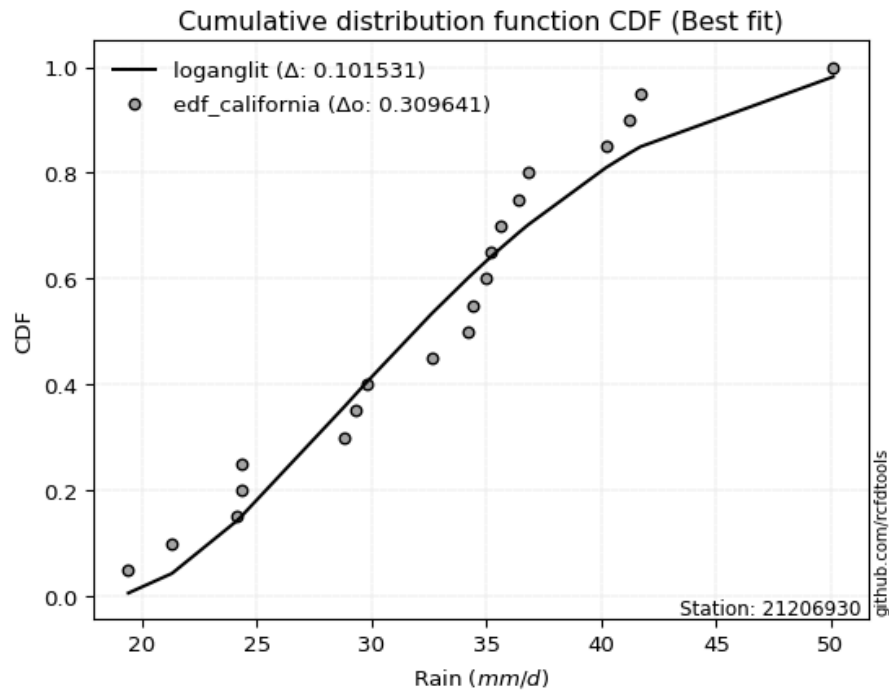
	0	1	2	3
station	21206930	21206930	21206930	21206930
empirical_dist	edf_california	edf_california	edf_california	edf_califc
p_dist	loganglit	logexponweib	logjohnsonsb	johnsons
delta	0.101531271147833	0.10225474751009744	0.10243321600110478	0.102579
deltao	0.3096406529800002	0.3096406529800002	0.3096406529800002	0.309640
eval	Δo > Δ	Δo > Δ	Δo > Δ	Δo > Δ
fit	1	1	1	1
n	20	20	20	20

	0	1	2	3
best_fit	1	0	0	0
best_fit_sort	1	2	3	4

1.3. Best fit

id	station	empirical_dist	p_dist	delta	deltao	eval	fit	n	best_fit
0	21206930	edf_california	loganglit	0.101531	0.309641	$\Delta o > \Delta$	1	20	1





2. EDF Hazen (1930)

Hazen method for plotting positions is a formula used to estimate the empirical cumulative probability distribution of flood events or other hydrological data. This formula often results in biased estimations, particularly when extrapolating to extreme events (high return periods).

$$P = (m - 0.5) / n$$

2.1. Empirical values

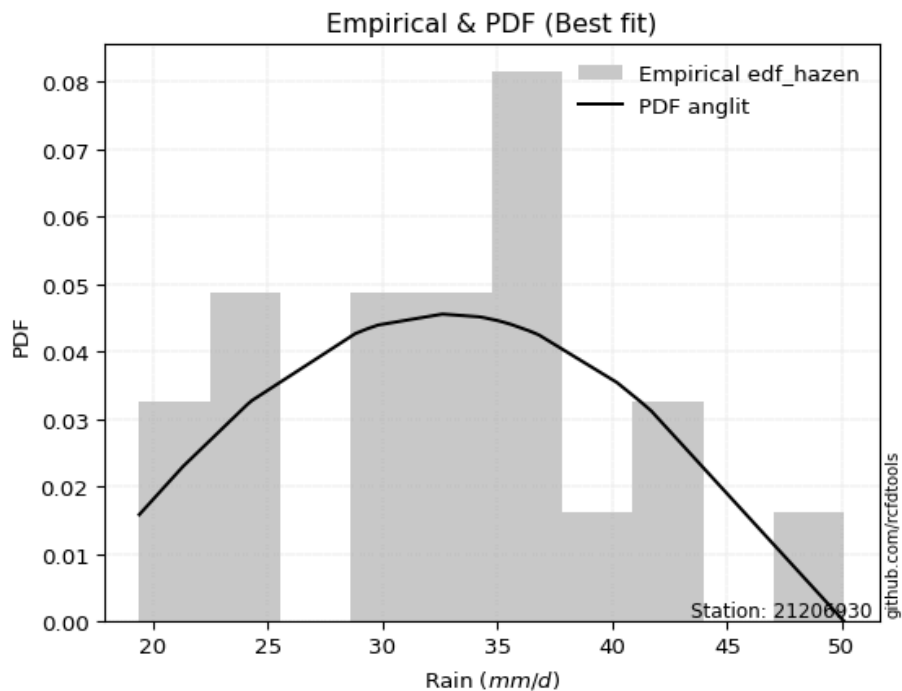
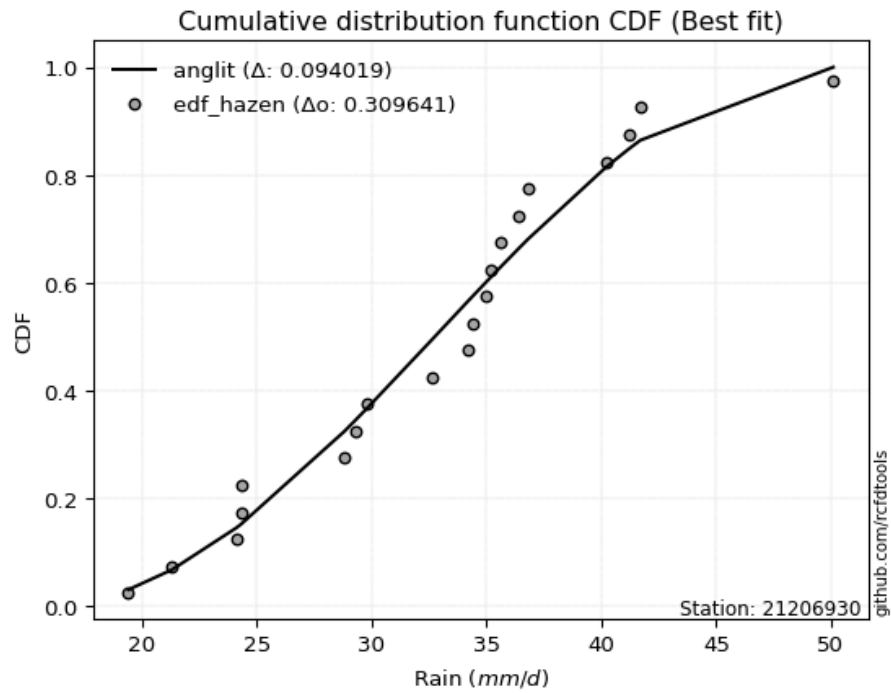
	0	1	2	3
date	2009	2025	2007	2018
x	19.4	21.3	24.1	24.3
m	1	2	3	4
empirical_dist	edf_hazen	edf_hazen	edf_hazen	edf_hazen
empirical	0.025	0.075	0.125	0.175
empirical_tr	1.0256410256410258	1.081081081081081	1.1428571428571428	1.212121212

2.2. Kolmogorov-Smirnov fit test (sorted by Δ)

	0	1	2	3
station	21206930	21206930	21206930	21206930
empirical_dist	edf_hazen	edf_hazen	edf_hazen	edf_hazen
p_dist	anglit	logburr	logmielke	lognormal
delta	0.09401863094548457	0.09605392674577629	0.09609571231859287	0.09760
deltao	0.3096406529800002	0.3096406529800002	0.3096406529800002	0.30964
eval	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$
fit	1	1	1	1
n	20	20	20	20
best_fit	1	0	0	0
best_fit_sort	1	2	3	4

2.3. Best fit

id	station	empirical_dist	p_dist	delta	deltao	eval	fit	n	best_fit
0	21206930	edf_hazen	anglit	0.0940186	0.309641	$\Delta_o > \Delta$	1	20	1



3. EDF Weibull (1939)

Weibull plotting position formula is an empirical method used to estimate the non-exceedance probability or plotting position for a set of observed data, is often recommended or widely used in practice, particularly in flood frequency analysis.

$$P = m / (n + 1)$$

3.1. Empirical values

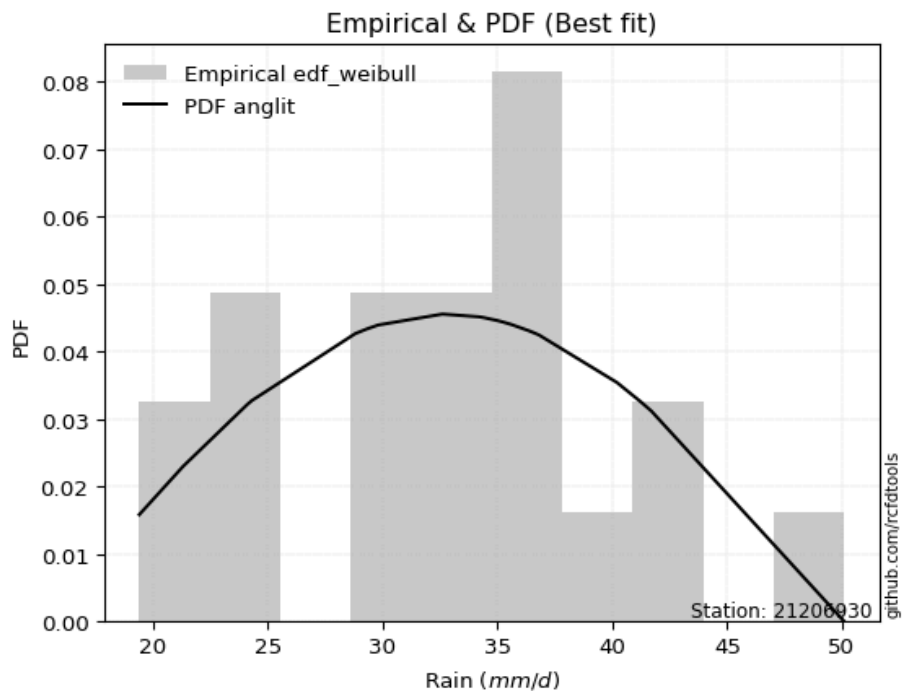
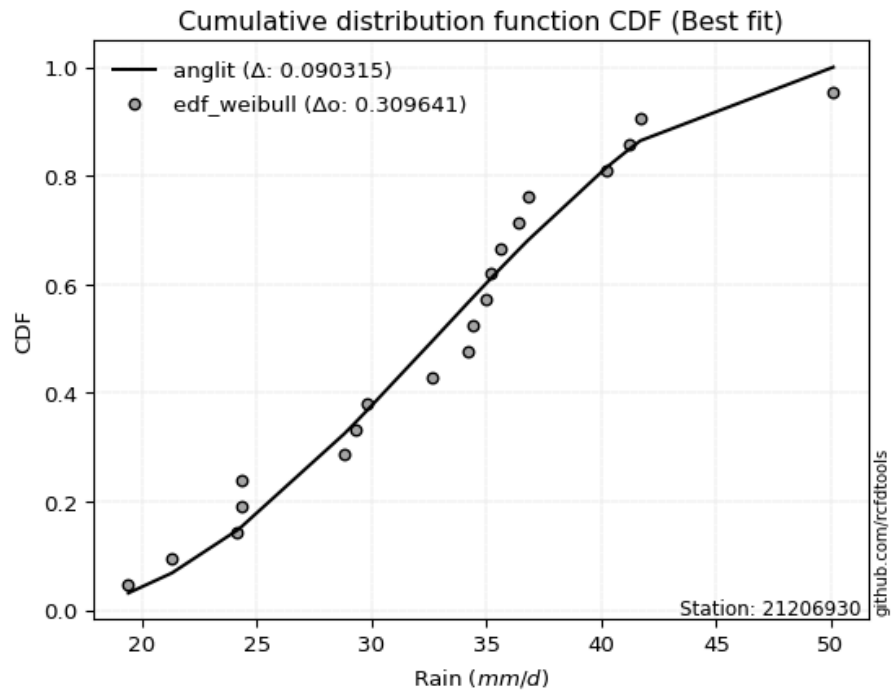
	0	1	2	3
date	2009	2025	2007	2018
x	19.4	21.3	24.1	24.3
m	1	2	3	4
empirical_dist	edf_weibull	edf_weibull	edf_weibull	edf_we
empirical	0.047619047619047616	0.09523809523809523	0.14285714285714285	0.1904
empirical_tr	1.05	1.1052631578947367	1.1666666666666665	1.2352

3.2. Kolmogorov-Smirnov fit test (sorted by Δ)

	0	1	2	3
station	21206930	21206930	21206930	21206930
empirical_dist	edf_weibull	edf_weibull	edf_weibull	edf_weil
p_dist	anglit	semicircular	gennorm	logtrunc
delta	0.09031541335958759	0.09159472639534316	0.10087017462072462	0.10128
deltao	0.3096406529800002	0.3096406529800002	0.3096406529800002	0.30964
eval	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$
fit	1	1	1	1
n	20	20	20	20
best_fit	1	0	0	0
best_fit_sort	1	2	3	4

3.3. Best fit

id	station	empirical_dist	p_dist	delta	deltao	eval	fit	n	best_fit
0	21206930	edf_weibull	anglit	0.0903154	0.309641	$\Delta_o > \Delta$	1	20	1



4. EDF Beard (1943)

The Beard formula (or Beard's plotting position formula) in hydrology is used to estimate the empirical non-exceedance probability (P) of a flood event (or other extreme hydrological data point) within a given dataset.

$$P = (m - 0.31) / (n + 0.38)$$

4.1. Empirical values

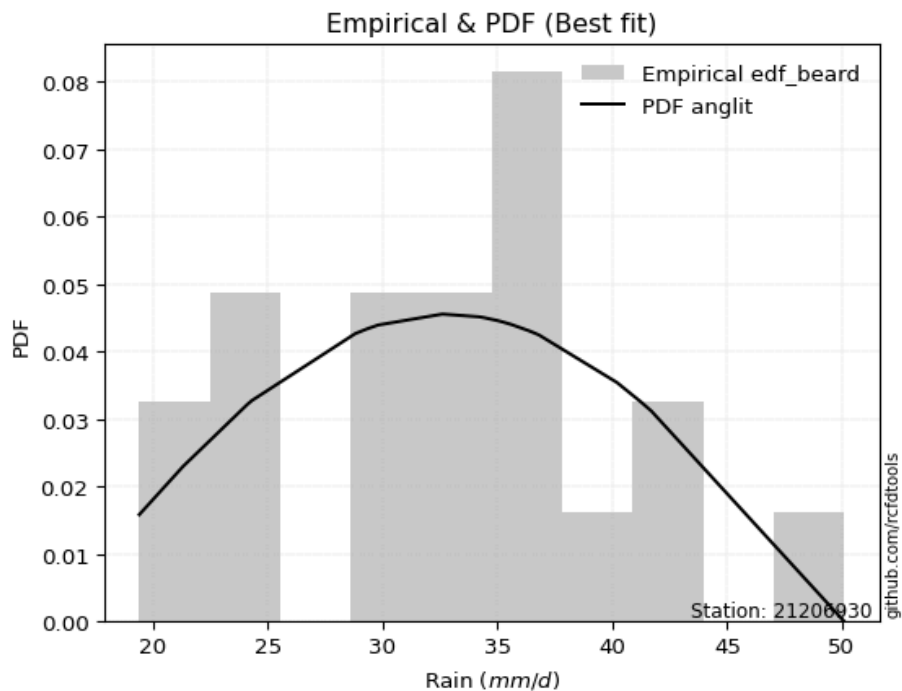
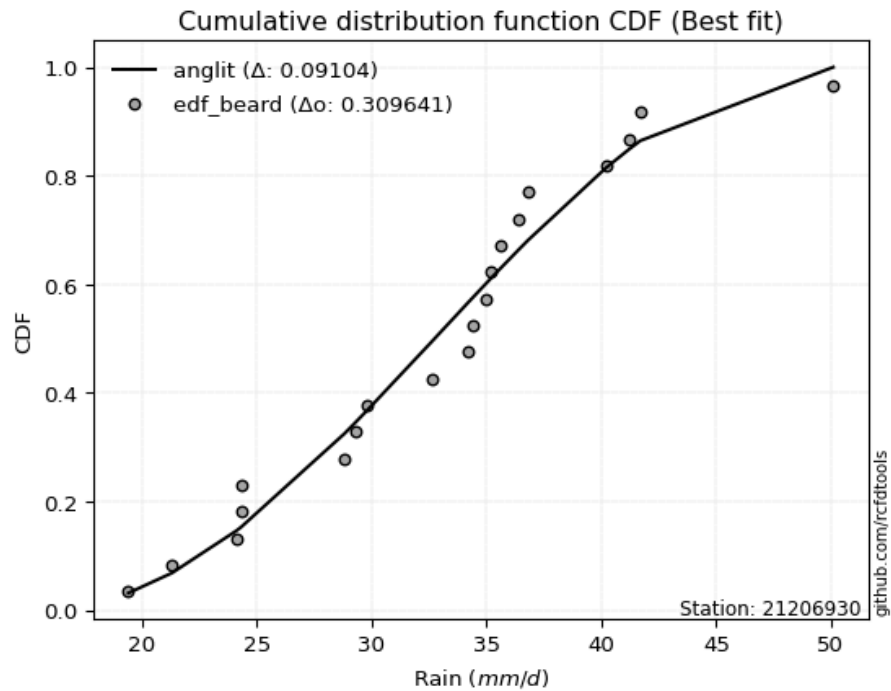
	0	1	2	3
date	2009	2025	2007	2018
x	19.4	21.3	24.1	24.3
m	1	2	3	4
empirical_dist	edf_beard	edf_beard	edf_beard	edf_be
empirical	0.033856722276741906	0.08292443572129539	0.13199214916584887	0.1810
empirical_tr	1.0350431691213813	1.090422685928304	1.152063312605992	1.2210

4.2. Kolmogorov-Smirnov fit test (sorted by Δ)

	0	1	2	3
station	21206930	21206930	21206930	21206930
empirical_dist	edf_beard	edf_beard	edf_beard	edf_beard
p_dist	anglit	semicircular	logburr	gennorr
delta	0.09103974627234052	0.09956238843562548	0.10118150280073215	0.10120
deltao	0.3096406529800002	0.3096406529800002	0.3096406529800002	0.30964
eval	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$
fit	1	1	1	1
n	20	20	20	20
best_fit	1	0	0	0
best_fit_sort	1	2	3	4

4.3. Best fit

id	station	empirical_dist	p_dist	delta	deltao	eval	fit	n	best_fit
0	21206930	edf_beard	anglit	0.0910397	0.309641	$\Delta_o > \Delta$	1	20	1



5. EDF Chegodayev (1955)

The Chegodayev formula is an empirical plotting position formula used in hydrological frequency analysis to estimate the exceedance probability or return period of a specific event from a set of observed data. It is primarily used for plotting observed data points on probability paper to fit a theoretical distribution, particularly for analyzing extreme events like maximum flood flows or rainfall intensities. The constant b value in the generalized plotting position formula is 0.3.

$$P = (m - b) / (n + 1 - 2b)$$

5.1. Empirical values

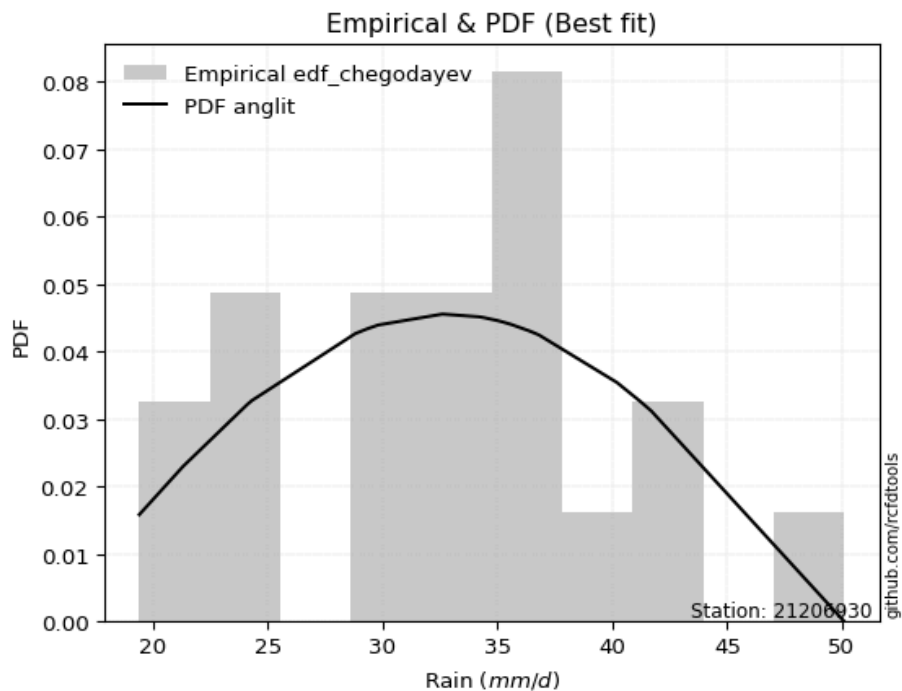
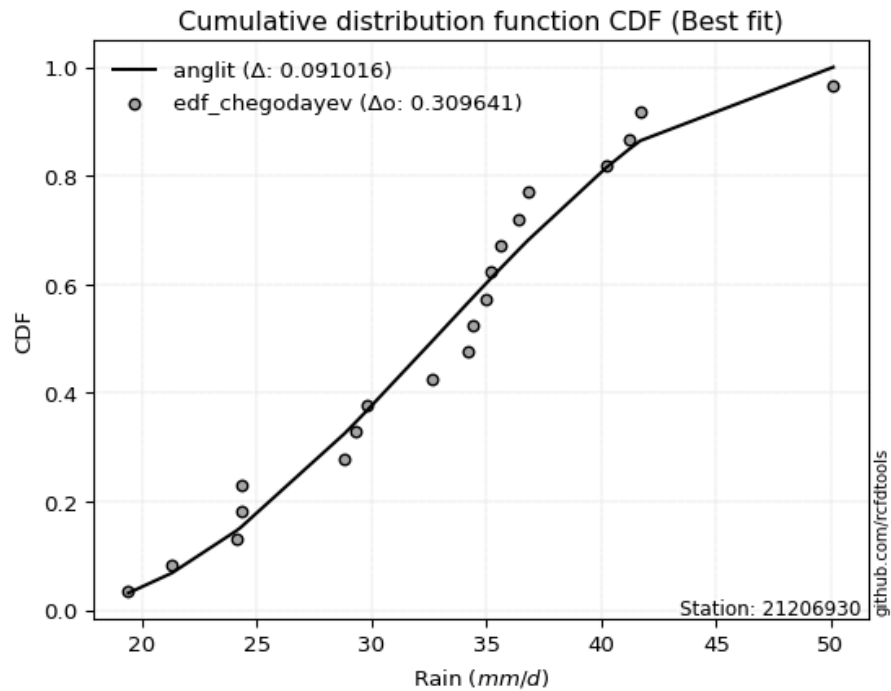
	0	1	2	3
date	2009	2025	2007	2018
x	19.4	21.3	24.1	24.3
m	1	2	3	4
empirical_dist	edf_chegodayev	edf_chegodayev	edf_chegodayev	edf_chegodayev
empirical	0.03431372549019608	0.08333333333333334	0.13235294117647062	0.18137
empirical_tr	1.0355329949238579	1.090909090909091	1.152542372881356	1.22155

5.2. Kolmogorov-Smirnov fit test (sorted by Δ)

	0	1	2	3
station	21206930	21206930	21206930	21206930
empirical_dist	edf_chegodayev	edf_chegodayev	edf_chegodayev	edf_chegodayev
p_dist	anglit	semicircular	gennorm	logburr
delta	0.09101569347163241	0.09929780762783624	0.10118205816768155	0.10144
deltao	0.3096406529800002	0.3096406529800002	0.3096406529800002	0.30964
eval	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$
fit	1	1	1	1
n	20	20	20	20
best_fit	1	0	0	0
best_fit_sort	1	2	3	4

5.3. Best fit

id	station	empirical_dist	p_dist	delta	deltao	eval	fit	n	best_fit
0	21206930	edf_chegodayev	anglit	0.0910157	0.309641	$\Delta_o > \Delta$	1	20	1



6. EDF Blom (1958)

The Blom formula is a specific "plotting position" formula used in hydrology and statistical analysis to estimate the empirical cumulative probability (or non-exceedance probability) of a data series. It is particularly recommended for data that are approximately normally distributed. The constant a is set to 0.375 (or $3/8$).

$$P = (m - a) / (n + 1 - 2a)$$

6.1. Empirical values

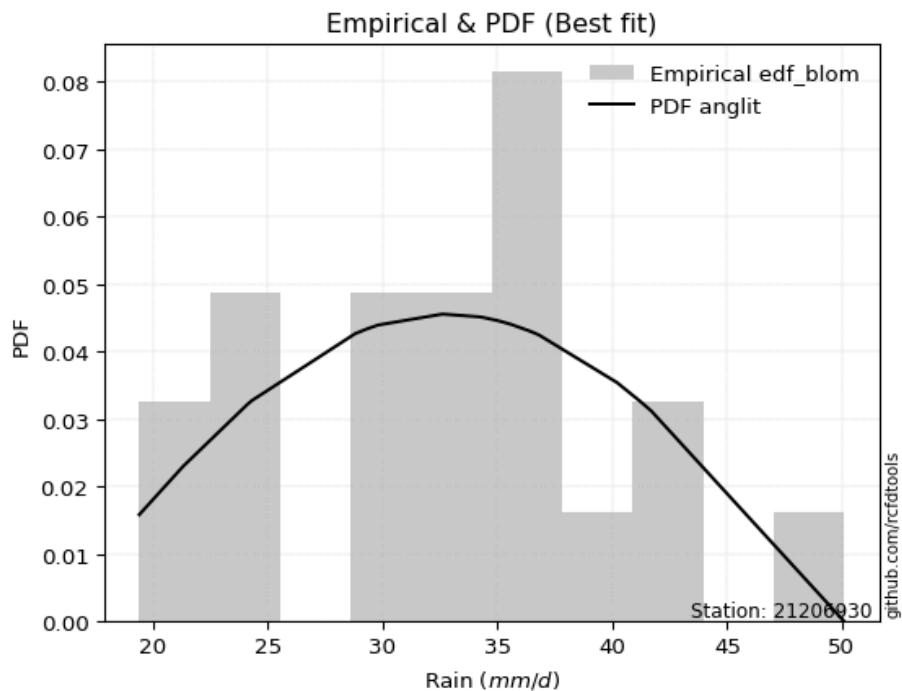
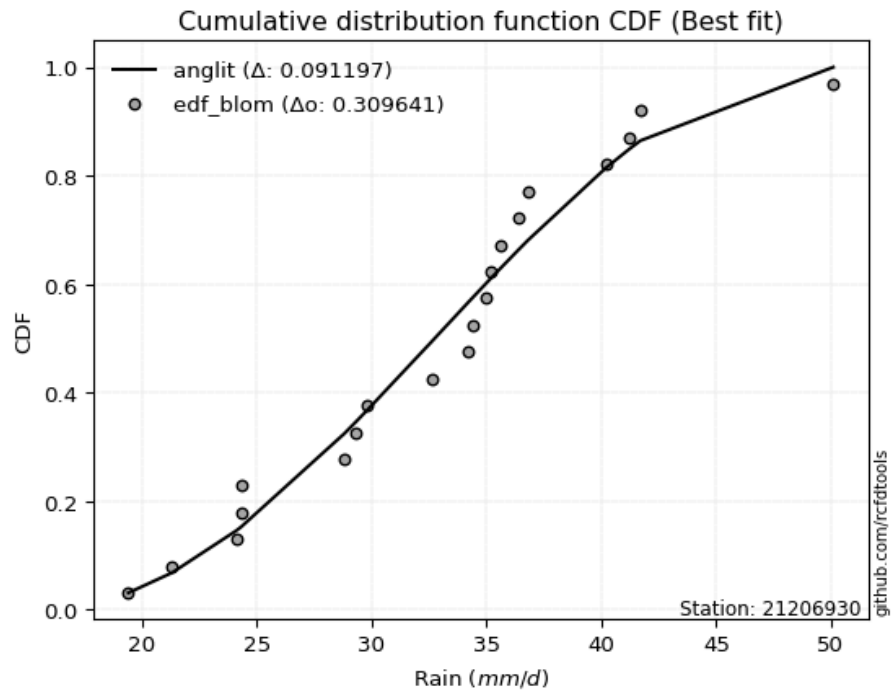
	0	1	2	3
date	2009	2025	2007	2018
x	19.4	21.3	24.1	24.3
m	1	2	3	4
empirical_dist	edf_blom	edf_blom	edf_blom	edf_blom
empirical	0.030864197530864196	0.08024691358024691	0.12962962962962962	0.1790
empirical_tr	1.0318471337579618	1.087248322147651	1.148936170212766	1.2180

6.2. Kolmogorov-Smirnov fit test (sorted by Δ)

	0	1	2	3
station	21206930	21206930	21206930	21206930
empirical_dist	edf_blom	edf_blom	edf_blom	edf_blom
p_dist	anglit	logburr	logmielke	loggenl
delta	0.09119724757475511	0.09944898847417133	0.09949077404698792	0.10100
deltao	0.3096406529800002	0.3096406529800002	0.3096406529800002	0.30964
eval	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$
fit	1	1	1	1
n	20	20	20	20
best_fit	1	0	0	0
best_fit_sort	1	2	3	4

6.3. Best fit

id	station	empirical_dist	p_dist	delta	deltao	eval	fit	n	best_fit
0	21206930	edf_blom	anglit	0.0911972	0.309641	$\Delta_o > \Delta$	1	20	1



7. EDF Tukey (1962)

In hydrology, the Tukey formula is used as a plotting position formula to estimate the empirical probability or frequency of a flood event (or other hydrological data). The formula parameter is given as $c=0.333$ (or $1/3$).

$$P = (m - c) / (n + 1 - 2c)$$

7.1. Empirical values

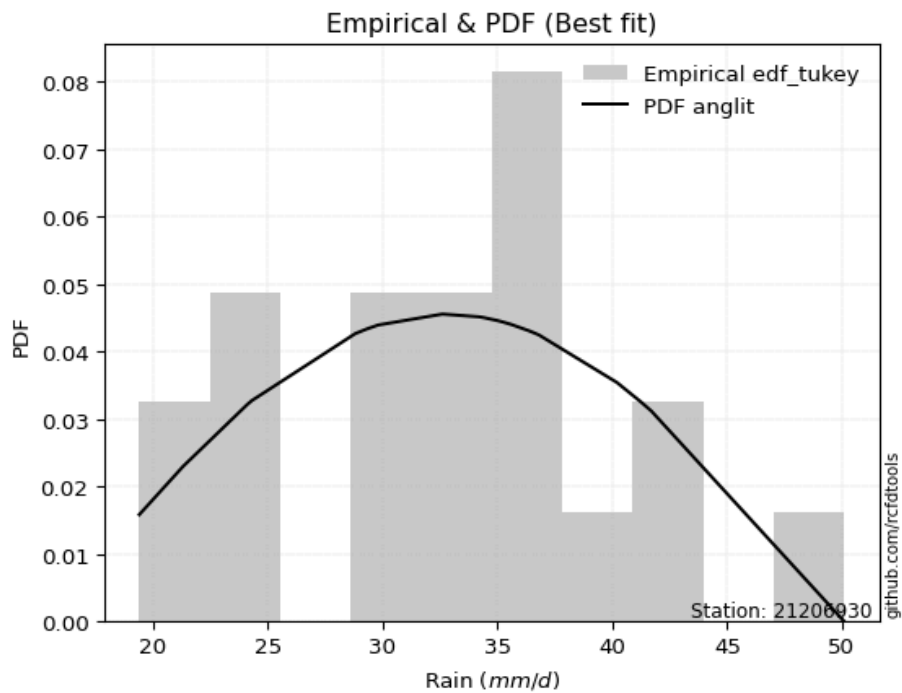
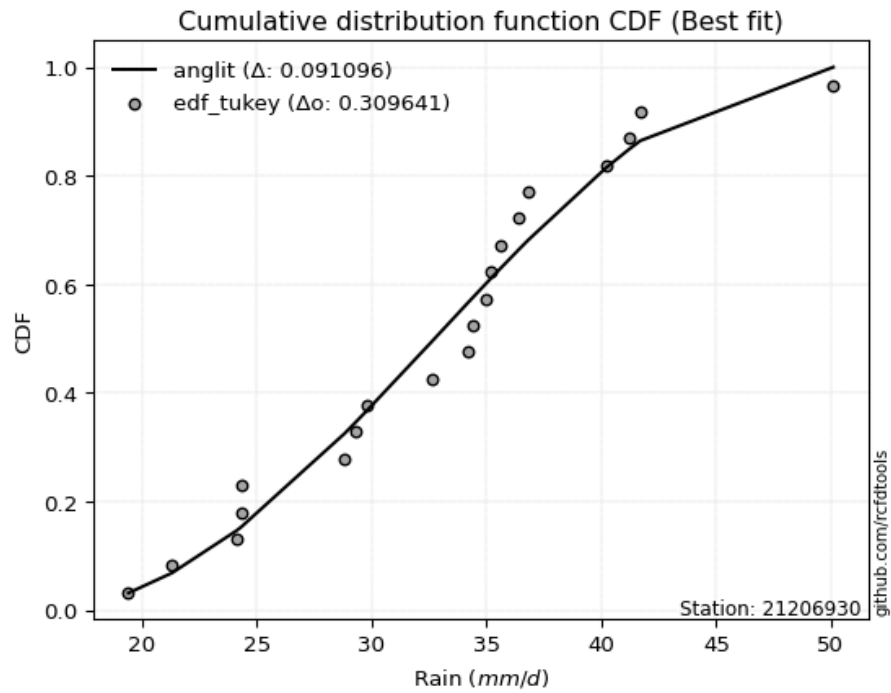
	0	1	2	3
date	2009	2025	2007	2018
x	19.4	21.3	24.1	24.3
m	1	2	3	4
empirical_dist	edf_tukey	edf_tukey	edf_tukey	edf_tukey
empirical	0.03278688524590164	0.08196721311475409	0.13114754098360656	0.18032
empirical_tr	1.033898305084746	1.0892857142857142	1.150943396226415	1.22

7.2. Kolmogorov-Smirnov fit test (sorted by Δ)

	0	1	2	3
station	21206930	21206930	21206930	21206930
empirical_dist	edf_tukey	edf_tukey	edf_tukey	edf_tukey
p_dist	anglit	semicircular	logburr	logmiell
delta	0.09109605348449001	0.10018176776926979	0.10056212346708776	0.10060
deltao	0.3096406529800002	0.3096406529800002	0.3096406529800002	0.30964
eval	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$
fit	1	1	1	1
n	20	20	20	20
best_fit	1	0	0	0
best_fit_sort	1	2	3	4

7.3. Best fit

id	station	empirical_dist	p_dist	delta	deltao	eval	fit	n	best_fit
0	21206930	edf_tukey	anglit	0.0910961	0.309641	$\Delta_o > \Delta$	1	20	1



8. EDF Gringorten (1963)

Gringorten plotting position formula is essential for estimating the probability and return periods of extreme events like floods and heavy rainfall. The constant $a=0.44$.

$$P = (m - a) / (n + 1 - 2a)$$

8.1. Empirical values

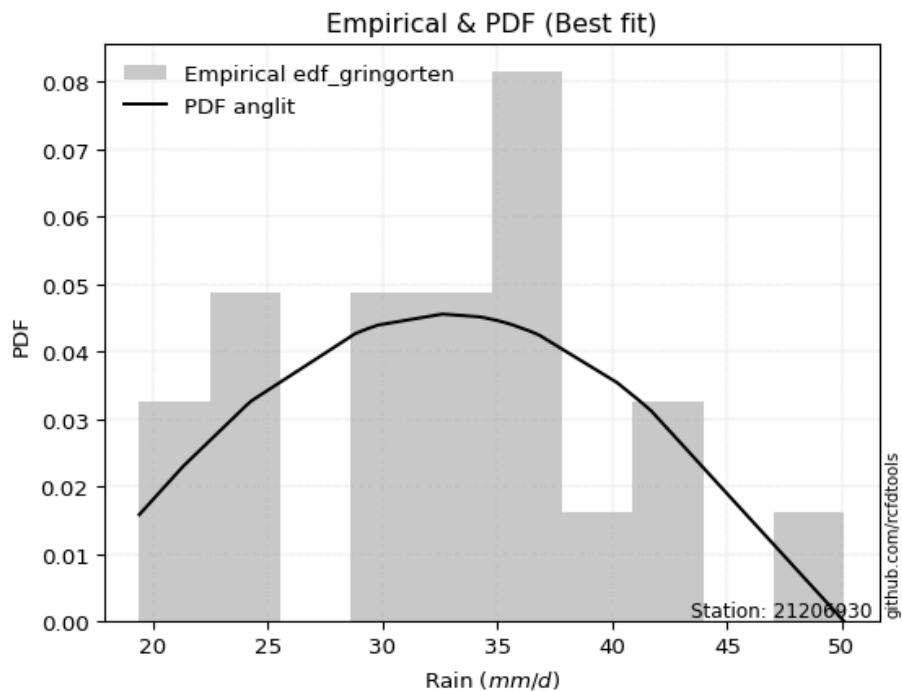
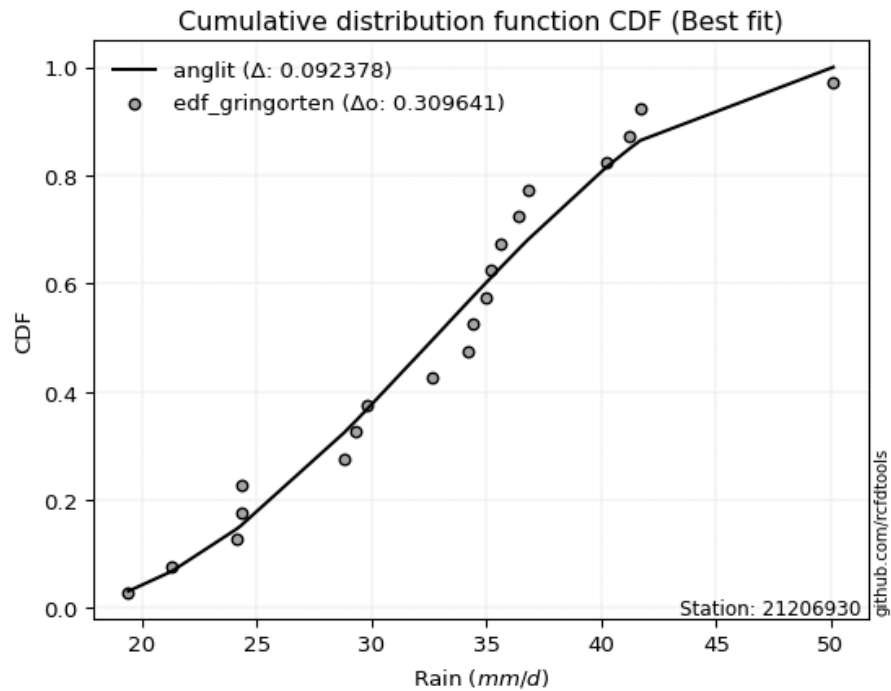
	0	1	2	3
date	2009	2025	2007	2018
x	19.4	21.3	24.1	24.3
m	1	2	3	4
empirical_dist	edf_gringorten	edf_gringorten	edf_gringorten	edf_gringorten
empirical	0.027833001988071572	0.0775347912524851	0.1272365805168986	0.176938
empirical_tr	1.0286298568507157	1.084051724137931	1.1457858769931661	1.214975

8.2. Kolmogorov-Smirnov fit test (sorted by Δ)

	0	1	2	3
station	21206930	21206930	21206930	21206930
empirical_dist	edf_gringorten	edf_gringorten	edf_gringorten	edf_gringorten
p_dist	anglit	logburr	logmielke	loggenlogis
delta	0.0923784718997589	0.0976940857915019	0.0977358713643185	0.09924977
deltao	0.3096406529800002	0.3096406529800002	0.3096406529800002	0.30964065
eval	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$
fit	1	1	1	1
n	20	20	20	20
best_fit	1	0	0	0
best_fit_sort	1	2	3	4

8.3. Best fit

id	station	empirical_dist	p_dist	delta	deltao	eval	fit	n	best_fit
0	21206930	edf_gringorten	anglit	0.0923785	0.309641	$\Delta_o > \Delta$	1	20	1



9. EDF Filliben (1975)

The specific values of the constants (0.3175) (often denoted as *alpha*) and (0.365) are derived from a method proposed by James J. Filliben in a 1975 paper. This particular formula is the mean value of the *i*-th order statistic of the normal distribution and is considered a robust and effective plotting position formula for the normal probability plot correlation coefficient test for normality.

$$P = (m - 0.3175) / (n + 0.365)$$

9.1. Empirical values

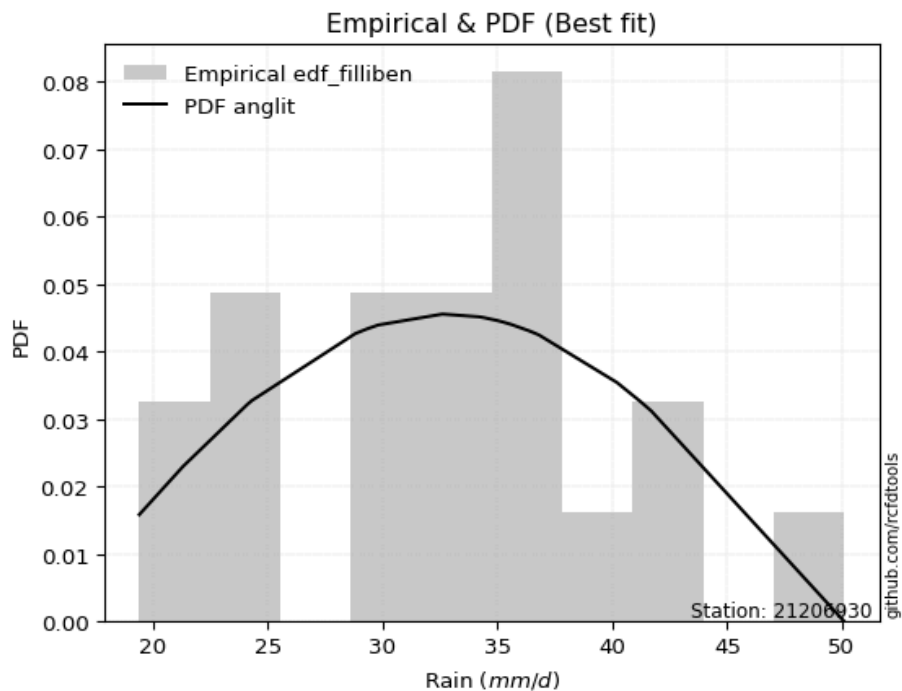
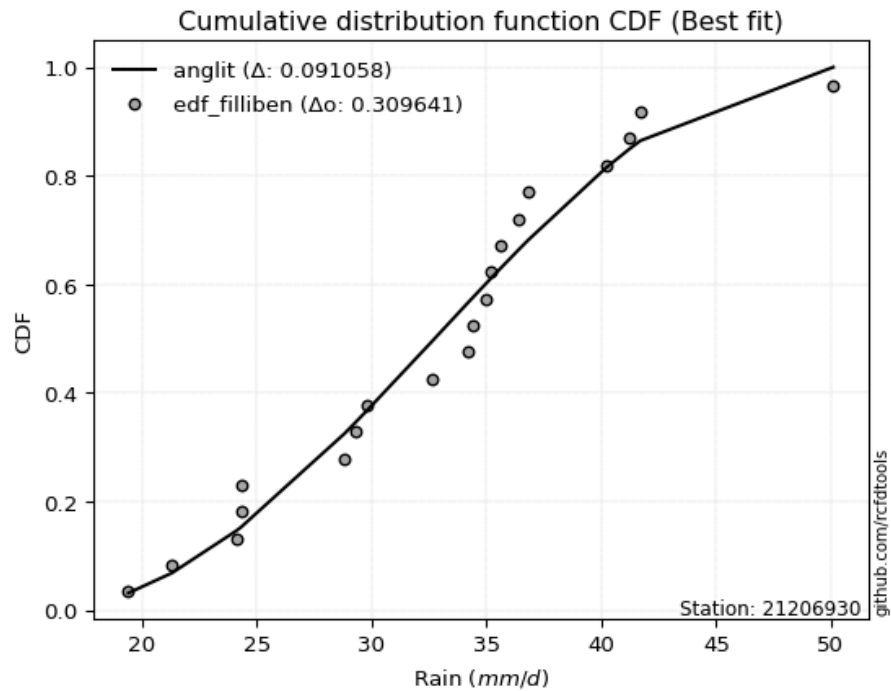
	0	1	2	3
date	2009	2025	2007	2018
x	19.4	21.3	24.1	24.3
m	1	2	3	4
empirical_dist	edf_filliben	edf_filliben	edf_filliben	edf_fillibe
empirical	0.03351338080039283	0.08261723545298306	0.1317210901055733	0.180824
empirical_tr	1.0346754731360346	1.090057540479058	1.1517036618125265	1.220740

9.2. Kolmogorov-Smirnov fit test (sorted by Δ)

	0	1	2	3
station	21206930	21206930	21206930	21206930
empirical_dist	edf_filliben	edf_filliben	edf_filliben	edf_fillik
p_dist	anglit	semicircular	logburr	logmiell
delta	0.09105781687635889	0.09976116507982757	0.10098272615653003	0.10102
deltao	0.3096406529800002	0.3096406529800002	0.3096406529800002	0.30964
eval	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$
fit	1	1	1	1
n	20	20	20	20
best_fit	1	0	0	0
best_fit_sort	1	2	3	4

9.3. Best fit

id	station	empirical_dist	p_dist	delta	deltao	eval	fit	n	best_fit
0	21206930	edf_filliben	anglit	0.0910578	0.309641	$\Delta_o > \Delta$	1	20	1



10. EDF Jenkinson (1977)

The Jenkinson formula in hydrology is an empirical plotting position formula used to estimate the non-exceedance probability (P) or return period (T) of a given ordered observation within a sample. It is a widely used method in the frequency analysis of extreme events such as floods and rainfall, as it provides a distribution-free way to plot data. $a \approx 0.31$ and $b \approx 0.38$ are constants derived to approximate the median of the probability distribution for the given rank.

$$P = (m - a) / (n + b)$$

10.1. Empirical values

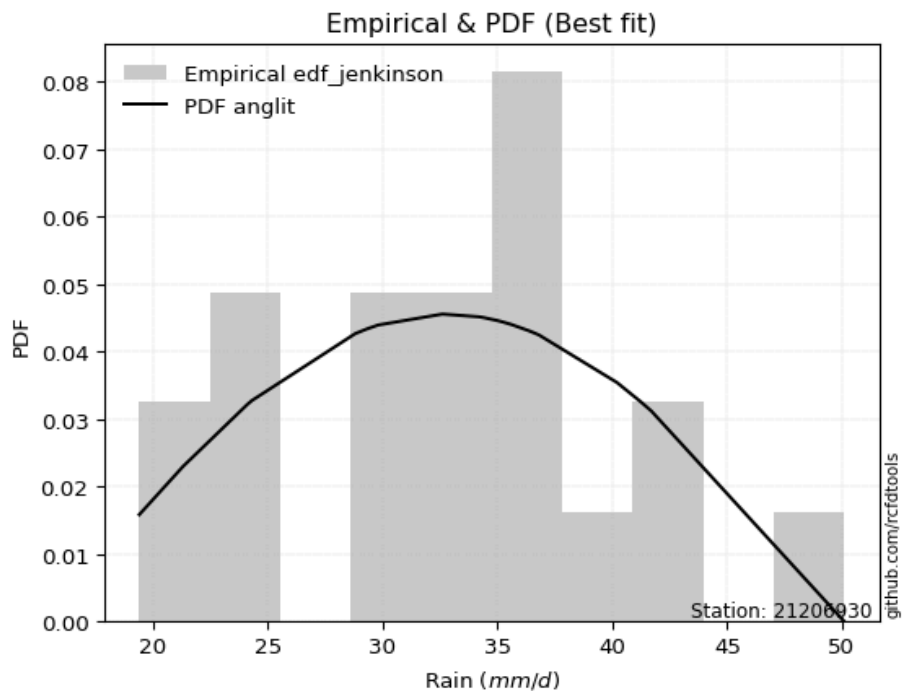
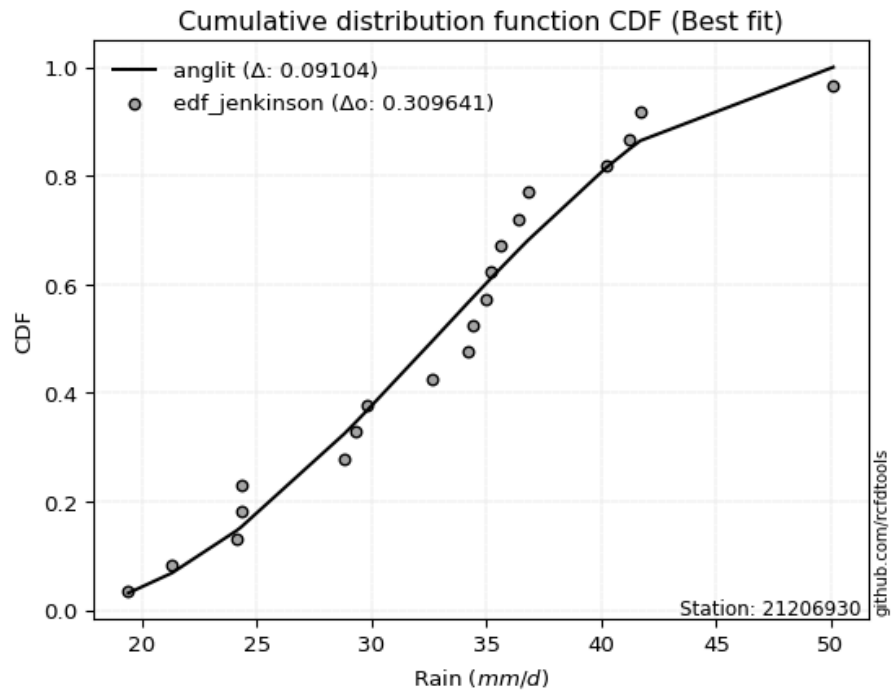
	0	1	2	3
date	2009	2025	2007	2018
x	19.4	21.3	24.1	24.3
m	1	2	3	4
empirical_dist	edf_jenkinson	edf_jenkinson	edf_jenkinson	edf_jer
empirical	0.033856722276741906	0.08292443572129539	0.13199214916584887	0.1810
empirical_tr	1.0350431691213813	1.090422685928304	1.152063312605992	1.2210

10.2. Kolmogorov-Smirnov fit test (sorted by Δ)

	0	1	2	3
station	21206930	21206930	21206930	21206930
empirical_dist	edf_jenkinson	edf_jenkinson	edf_jenkinson	edf_jenk
p_dist	anglit	semicircular	logburr	gennorr
delta	0.09103974627234052	0.09956238843562548	0.10118150280073215	0.10120
deltao	0.3096406529800002	0.3096406529800002	0.3096406529800002	0.30964
eval	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$
fit	1	1	1	1
n	20	20	20	20
best_fit	1	0	0	0
best_fit_sort	1	2	3	4

10.3. Best fit

id	station	empirical_dist	p_dist	delta	deltao	eval	fit	n	best_fit
0	21206930	edf_jenkinson	anglit	0.0910397	0.309641	$\Delta_o > \Delta$	1	20	1



11. EDF Cunnane (1978)

Cunnane's work in statistical hydrology has focused on the performance and evaluation of different probability distributions (such as GEV, Gumbel, Lognormal) for flood frequency estimation. b is a constant, typically set to 0.4.

$$P = (m - b) / (n + 1 - 2b)$$

11.1. Empirical values

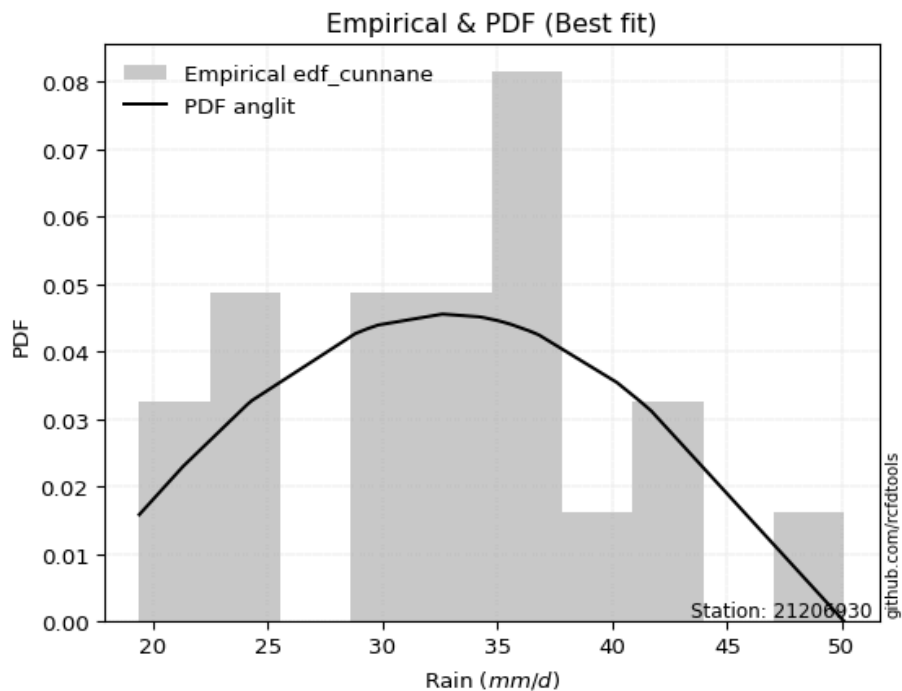
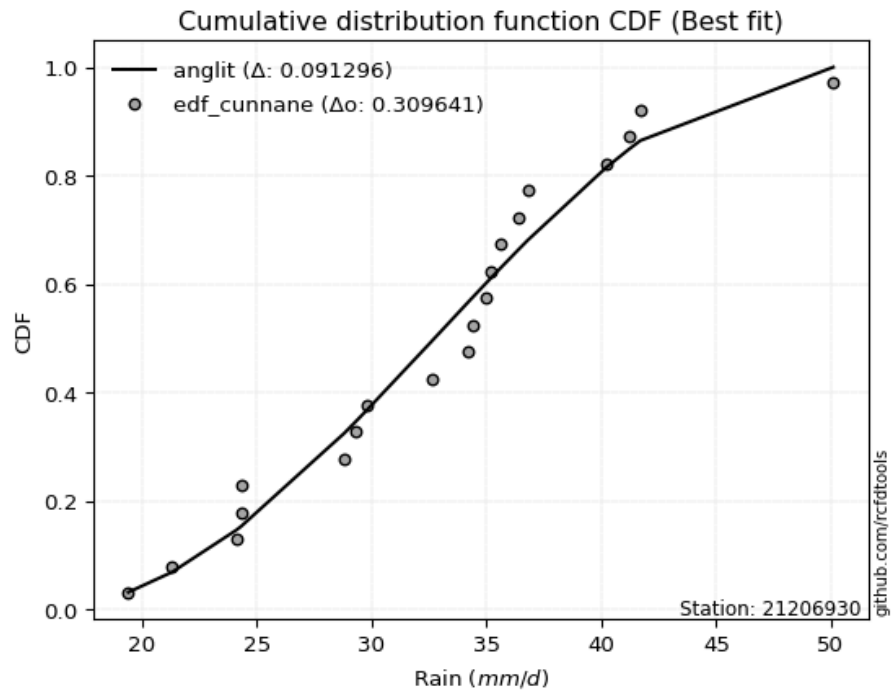
	0	1	2	3
date	2009	2025	2007	2018
x	19.4	21.3	24.1	24.3
m	1	2	3	4
empirical_dist	edf_cunnane	edf_cunnane	edf_cunnane	edf_cunn
empirical	0.0297029702970297	0.07920792079207921	0.12871287128712872	0.178217
empirical_tr	1.030612244897959	1.086021505376344	1.1477272727272727	1.216867

11.2. Kolmogorov-Smirnov fit test (sorted by Δ)

	0	1	2	3
station	21206930	21206930	21206930	21206930
empirical_dist	edf_cunnane	edf_cunnane	edf_cunnane	edf_cunn
p_dist	anglit	logburr	logmielke	loggenl
delta	0.09129585866825685	0.09877669902300398	0.09881848459582057	0.10033
deltao	0.3096406529800002	0.3096406529800002	0.3096406529800002	0.30964
eval	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$
fit	1	1	1	1
n	20	20	20	20
best_fit	1	0	0	0
best_fit_sort	1	2	3	4

11.3. Best fit

id	station	empirical_dist	p_dist	delta	deltao	eval	fit	n	best_fit
0	21206930	edf_cunnane	anglit	0.0912959	0.309641	$\Delta_o > \Delta$	1	20	1



12. EDF Adamowski (1981)

The Adamowski formula in hydrology refers to a specific plotting position formula used for estimating the non-parametric empirical distribution of hydrological events (like flood peaks) to calculate their return periods. This formula provides an alternative to traditional parametric methods (like the Gumbel or Log Pearson Type III distributions).

$$P = (m - 0.25) / (n + 0.5)$$

12.1. Empirical values

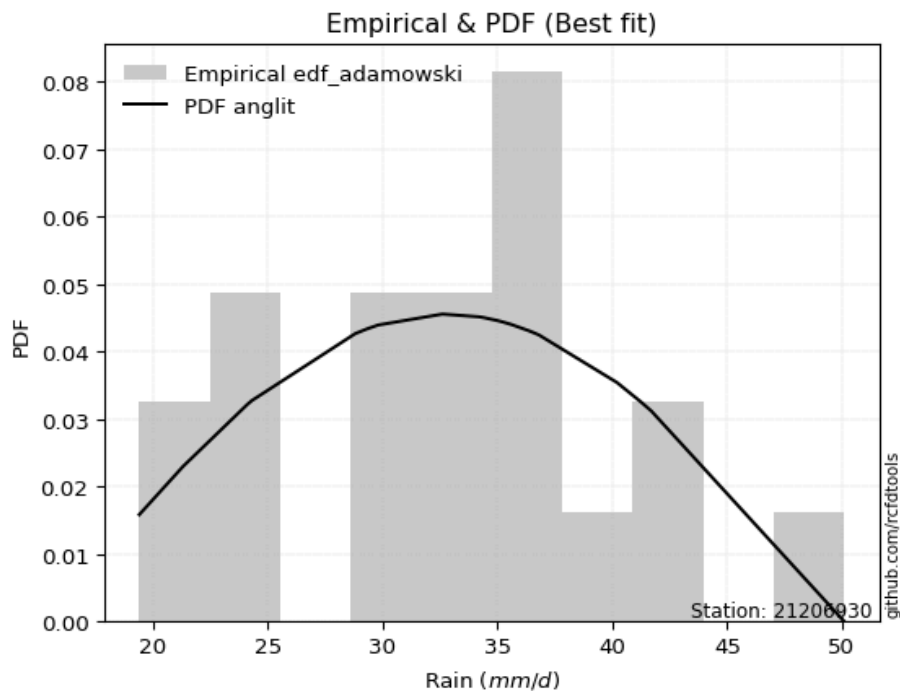
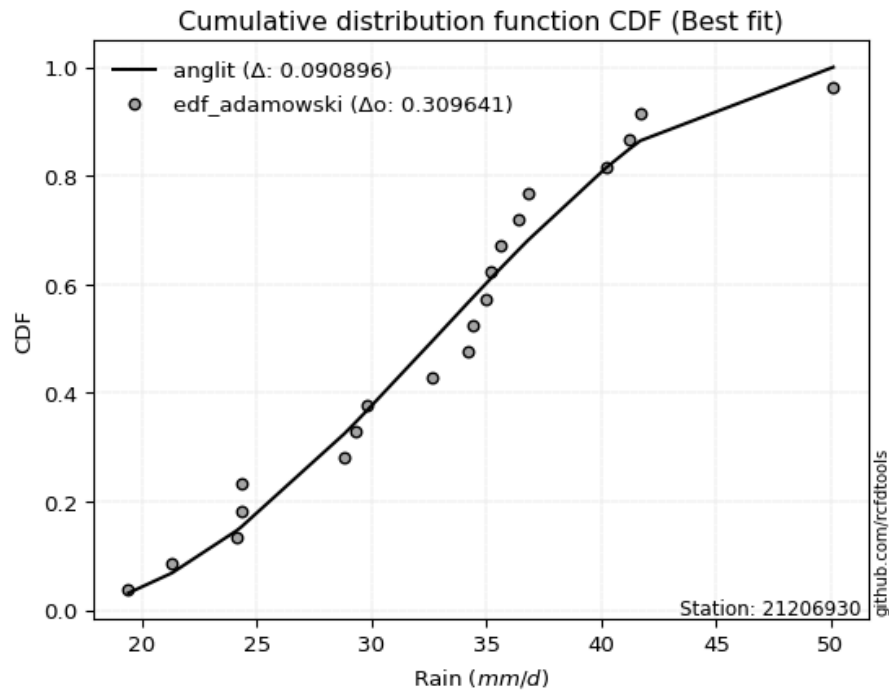
	0	1	2	3
date	2009	2025	2007	2018
x	19.4	21.3	24.1	24.3
m	1	2	3	4
empirical_dist	edf_adamowski	edf_adamowski	edf_adamowski	edf_ad
empirical	0.036585365853658534	0.08536585365853659	0.13414634146341464	0.1829
empirical_tr	1.0379746835443038	1.0933333333333333	1.1549295774647887	1.2238

12.2. Kolmogorov-Smirnov fit test (sorted by Δ)

	0	1	2	3
station	21206930	21206930	21206930	21206930
empirical_dist	edf_adamowski	edf_adamowski	edf_adamowski	edf_adamowski
p_dist	anglit	semicircular	gennorm	t
delta	0.0908961334525028	0.09798264741741058	0.10106249814855195	0.101726
deltao	0.3096406529800002	0.3096406529800002	0.3096406529800002	0.309640
eval	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$	$\Delta_o > \Delta$
fit	1	1	1	1
n	20	20	20	20
best_fit	1	0	0	0
best_fit_sort	1	2	3	4

12.3. Best fit

id	station	empirical_dist	p_dist	delta	deltao	eval	fit	n	best_fit
0	21206930	edf_adamowski	anglit	0.0908961	0.309641	$\Delta_o > \Delta$	1	20	1



C. Best fit & Estimate extreme values for specific return periods - Tr

Best fit in probability distribution analysis means finding the theoretical distribution (like Normal, Poisson, Exponential, etc.) that most accurately mirrors your real-world data's patterns (shape, center, spread) for better predictions, using methods like visual checks (Q-Q plots), goodness-of-fit tests (Kolmogorov-Smirnov, Chi-Squared), and information criteria (AIC) to score and select the simplest, most representative model.

1. Best fit (ordered by delta Δ)

id	station	empirical_dist	p_dist	delta	deltao	eval	fit	n	best
0	21206930	edf_weibull	anglit	0.0903154	0.309641	$\Delta o > \Delta$	1	20	
1	21206930	edf_adamowski	anglit	0.0908961	0.309641	$\Delta o > \Delta$	1	20	
2	21206930	edf_chegodayev	anglit	0.0910157	0.309641	$\Delta o > \Delta$	1	20	
3	21206930	edf_jenkinson	anglit	0.0910397	0.309641	$\Delta o > \Delta$	1	20	
4	21206930	edf_beard	anglit	0.0910397	0.309641	$\Delta o > \Delta$	1	20	
5	21206930	edf_filliben	anglit	0.0910578	0.309641	$\Delta o > \Delta$	1	20	
6	21206930	edf_tukey	anglit	0.0910961	0.309641	$\Delta o > \Delta$	1	20	
7	21206930	edf_blom	anglit	0.0911972	0.309641	$\Delta o > \Delta$	1	20	
8	21206930	edf_cunnane	anglit	0.0912959	0.309641	$\Delta o > \Delta$	1	20	
9	21206930	edf_gringorten	anglit	0.0923785	0.309641	$\Delta o > \Delta$	1	20	
10	21206930	edf_hazen	anglit	0.0940186	0.309641	$\Delta o > \Delta$	1	20	
11	21206930	edf_california	loganglit	0.101531	0.309641	$\Delta o > \Delta$	1	20	

2. Extreme values and % difference analysis

In hydrology, a return period (or recurrence interval Tr) is the statistical average time between extreme events like floods or droughts of a specific magnitude, indicating how rare an event is, with a 100-year flood, for example, having a 1% chance of occurring in any given year, not that it happens exactly every century. It is a key tool for infrastructure design (like bridges or dams) and risk assessment, calculated from historical data to determine the probability of future events, although it is important to remember it is statistical average, and events can cluster or be missed.

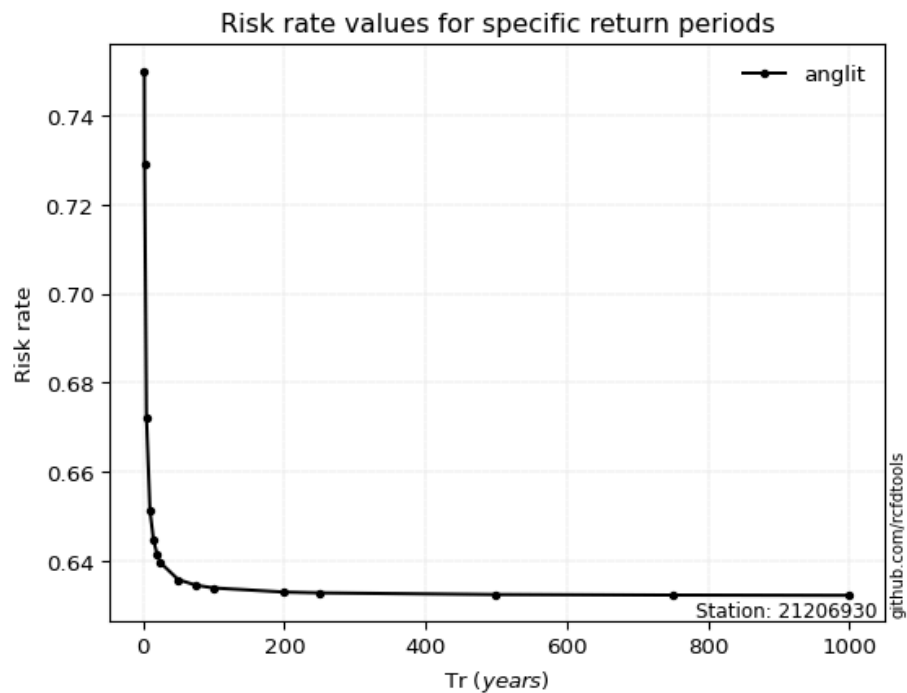
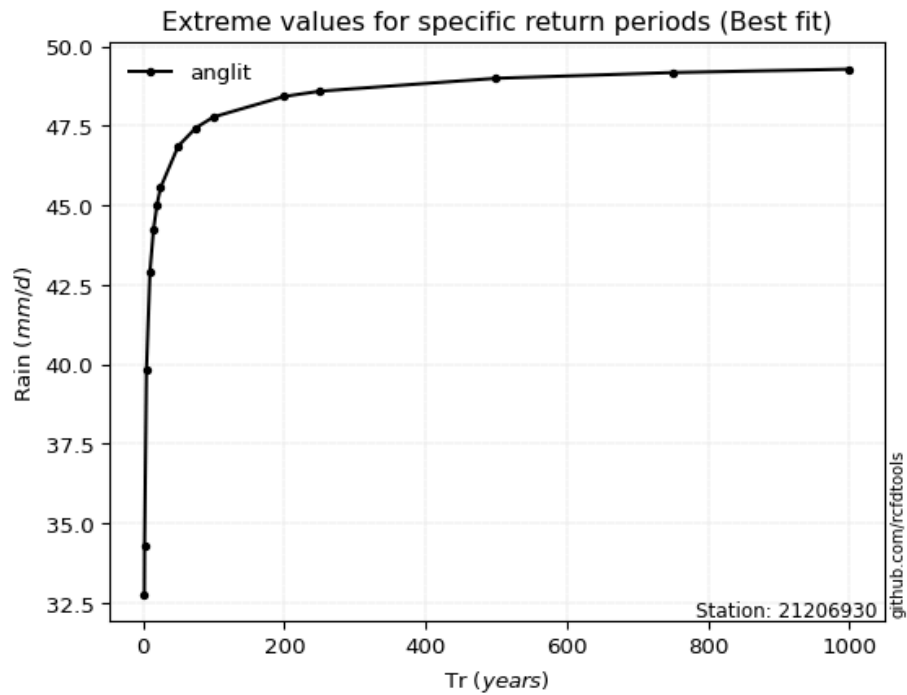
The extreme value percentage difference is evaluated as $pdiff = 1 - (ETrBf / ETr) * 100$, where $ETrBf$ correspond to the bestfit extreme value for a specific recurrence interval and ETr correspond to the extreme value for one of the most used PDFs used in hydrology.

risk_rate: assuming the return period as the project useful life.

- Table: [extreme_21206930.csv](#), all PDFs.
- Table: [extremepdiff_21206930.csv](#), % difference between bestfit and most used PDFs in Hydrology.

Extreme values table for only best fit PDFs

id	tr	prob_l	prob_g	alpha	anglit	arcsine	argus	beta	bei
0	2	0.5	0.5	32.3165	32.7362	32.7363	35.9923	32.4655	
1	2.33	0.570815	0.429185	33.6634	34.2955	35.0769	37.8014	34.8667	
2	5	0.8	0.2	38.889	39.7968	41.3186	43.6044	43.0267	
3	10	0.9	0.1	42.5602	42.9106	42.8254	46.4028	46.6651	
4	15	0.933333	0.0666667	44.4576	44.2403	43.1128	47.4666	47.8638	
5	20	0.95	0.05	45.7248	45.0224	43.214	48.0506	48.4554	
6	25	0.96	0.04	46.6704	45.5525	43.2609	48.4271	48.8065	
7	50	0.98	0.02	49.4401	46.8573	43.3236	49.2809	49.4945	
8	75	0.986667	0.0133333	50.9652	47.4316	43.3353	49.6195	49.7171	
9	100	0.99	0.01	52.0126	47.773	43.3393	49.8085	49.8263	
10	200	0.995	0.005	54.4376	48.4181	43.3433	50.1371	49.986	
11	250	0.996	0.004	55.1927	48.5823	43.3437	50.2139	50.017	
12	500	0.998	0.002	57.4713	48.9894	43.3444	50.3907	50.0776	
13	750	0.998667	0.00133333	58.7633	49.1697	43.3445	50.4617	50.0972	
14	1000	0.999	0.001	59.6641	49.2771	43.3445	50.5016	50.1068	



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